

A new multi-time-step method for implicit-explicit time integration of incompressible Navier-Stokes equations

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Abstract

Implicit-explicit (IMEX) time integration schemes have emerged as a robust numerical method for simulating problems in computation fluid dynamics. In particular, IMEX schemes in conjunction with the scalar auxiliary variable (SAV) approach are able to achieve unconditional energy stability. However, existing IMEX-SAV schemes are computationally intensive for problems with multiple temporal scales because they are constrained to use the smallest time step necessary to resolve the finest time scale in the entire fluid domain. In this paper, we develop a novel multi-time-step (MTS) method that enables the use of different time steps in different fluid subdomains while maintaining accuracy and unconditional stability. Subdomains with high temporal gradients are integrated with a small step to maintain accuracy whereas other subdomains are integrated with relatively larger time steps to achieve computational efficiency. Dual-Schur domain decomposition (DD) is used to divide the spatial problem domain into several subdomains and continuity of velocity and pressure is simultaneously enforced between the subdomains using Lagrange multipliers. Furthermore, a SAV approach is used to ensure unconditional stability of the method. We show that the method leads to a system of linear algebraic equations thereby eliminating the need for non-linear iterations and reducing the computational cost further. Finally, we demonstrate the performance of the proposed MTS-IMEX-SAV method and highlight the computational gains it achieves compared to conventional IMEX-SAV schemes by solving several benchmark problems.

Keywords: Incompressible Navier-Stokes equations, Time integration, Multi-time-step, Scalar auxiliary variable, Implicit-explicit, High-order BDF scheme

1. Introduction

In computational fluid dynamics (CFD) and fluid-structure interaction (FSI), there are three principal spatial discretization methods for solving Navier-Stokes equations: a) finite difference method (FDM), b) finite element method (FEM), and c) finite volume method (FVM) [1]. FDM is one of the earliest numerical approaches that were used for solving differential equations by discretizing the spatial domain and/or temporal domain with finite differences. FDM was already in use for CFD before the 1950s when the first instances of FEM were published. Historically, FDM has been used more than FEM for applications in CFD because of the simplicity of its formulation and the efficiency in computation. In recent years, FEM

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and FVM have become more common in CFD and FSI because discretizing complex domains with FDM can be challenging. FEM is frequently used to solve partial differential equations (PDEs), by dividing the problem domain into a number of small elements and approximating the solution fields over these elements with low-order polynomials. In contrast, with FVM, PDEs are represented in the form of integrals where the divergence term is evaluated using fluxes over each finite volume (or cell). Note that FVM was initially a derivative of the FDM and it was later shown that FVM can also be derived from FEM [1]. However, in FVM, it can be challenging to enforce constraints on meshes over problem domains with complex shapes [2]. On the other hand, even though the formulation of FEM can be more intricate, there are several advantages of FEM, such as allowing for fully unstructured and arbitrary grids, and providing more accurate numerical solutions to self-adjoint problems [3]. Recently, the discontinuous Galerkin (DG) method has also been used for CFD simulations [4–6]. The DG method combines features of both FEM and FVM in that the solution across element boundaries can be discontinuous and the coupling between elements is handled through numerical fluxes at the interfaces. Although the DG method can yield high-order accurate solutions, it generally leads to a higher computational cost compared to FEM and FVM. In this study, we focus on the FEM to discretize spatial domains for simulating fluid flows.

In conjunction with FEM, direct time integration schemes are widely used to solve non-linear ordinary differential equations (ODEs) in CFD and FSI problems [7, 8]. In current literature, time integration schemes are mainly divided into two classes, namely multi-step methods [9–13] and multi-stage methods [14–18]. Different methods in these two classes possess different numerical properties in terms of their stability, accuracy, and computational cost. Time integration schemes can further be categorized into explicit schemes [9, 10] and implicit schemes [11–13, 16]. Explicit schemes are commonly used for moving wave and shock problems because they have a low computational cost per time step and they are associated with a maximum critical time step given by the Courant-Friedrichs-Lewy (CFL) condition [19]. On the other hand, implicit schemes permit a greater range of time steps and several are unconditionally stable making them suitable for steady-state and incompressible flow problems. However, with implicit time integration schemes, there is a need for iterative algorithms, such as Picard iteration or Newton-Raphson iteration, for computing solutions to non-linear problems (e.g. the Navier-Stokes equations).

Implicit-explicit (IMEX) time integration schemes exploit the favorable characteristics of both, implicit and explicit schemes, by treating linear terms using implicit time integration and integrating non-linear terms with an explicit time discretization [20]. This avoids the need for non-linear iterations at each time step, keeping the computational cost low and leads to a larger domain of stability compared to the underlying explicit scheme. In some cases, however, an iterative solution is needed despite the use of IMEX schemes. For instance, Dumbser and Casulli [21] solved the compressible Navier-Stokes equations using an explicit operator for the non-linear convective terms and treating pressure implicitly in the momentum equations. This results in an implicit coupling between the momentum and energy equations which is solved using Picard iteration. Additionally, stability of IMEX schemes is not guaranteed for general non-linear problems

[22]. To overcome this limitation, in recent years, the scalar auxiliary variable (SAV) approach for gradient flows, introduced by Shen and his co-workers [23–26], has emerged as a robust method with high-order accuracy and energy stability. They formulated SAV schemes for several time stepping methods, such as the Crank-Nicolson method and the k th-order backward difference formula (BDF k) (where $k = 1, 2, \dots$). Lin et al. [27] proposed BDF k based on the SAV approach (BDF k SAV) to approximate incompressible Navier-Stokes equations. Shen and his co-workers have developed several SAV schemes for these equations as well: SAV schemes based on the Crank-Nicolson method [28]; IMEX BDF k SAV schemes [29]; SAV-pressure correction methods [30]; and IMEX BDF k SAV schemes with a consistent splitting scheme [31]. Recently, a SAV stabilization of IMEX schemes for non-linear structural dynamics has also been developed [20].

A key limitation of existing time integration schemes, including explicit, implicit, and IMEX schemes, is that almost all schemes are constrained to use a uniform time step (UTS) over the entire problem domain. For problems with multiple temporal scales, the UTS approach becomes rather inefficient because the computational cost of these schemes is governed by the smallest time step that is necessary to resolve the relevant fine scales. One way to achieve high overall accuracy at a low computational cost is to use the multi-time-step (MTS) approach in conjunction with domain decomposition (DD). In this strategy, the problem domain is divided into several subdomains, which are solved with different time stepping schemes and/or different time-step sizes that meet the stability and accuracy requirements of individual subdomains. The MTS approach is recognized as an efficient method for solving second-order ODEs, especially for problems in structural dynamics that exhibit multiple temporal scales [32–41]. The MTS strategy has been extended to problems governed by the first-order ODEs, such as transient heat-conduction [42–44].

In the current literature, to the best of the authors’ knowledge, there is no MTS method available for coupling IMEX-SAV schemes to solve incompressible Navier-Stokes equations. However, there are a few MTS-like techniques, such as local time stepping (LTS) and multi-rate time stepping (MRT), that have been formulated for different space and time discretizations. Osher and Richard [45] first proposed LTS approach for solving one-dimensional scalar conservation equation by using smaller time steps in high-resolution regions and larger time steps in low-resolution regions. Trahan and Dawson [46] developed a LTS method to solve shallow-water equations with non-overlapping domain decomposition approach, where the fast dynamics are advanced first, followed by the slow dynamics. Emmett et al. [47] solved multi-component reacting compressible Navier-Stokes equations by introducing a MRT strategy that enables the chemistry to be advanced on a time scale different from that of the fluid dynamics. Rybak et al. [48] proposed a MRT method for solving coupled porous medium and free flow systems by using small time scale for free flow and large time scale for porous medium. Gupta et al. [49] proposed two MRT schemes for partitioned multi-scale differential algebraic equations. Mikida et al. [50] presented a MRT approach using overlapping grids with explicit Adams-Bashforth time integration schemes, where the ODE corresponding to the faster dynamics is advanced first, followed by the ODE associated the slower dynamics. Mittal et al. [51] proposed a MRT method based on a stable overlapping Schwarz method for solving the incompressible Navier-Stokes

equations, independent of spatial discretization. In contrast, non-overlapping dual-Schur DD methods for Navier-Stokes equations, where Lagrange multipliers are used to enforce continuity of the solution across the interface between subdomains, have been restricted thus far to the UTS approach [52–57].

In this paper, we formulate a novel unconditionally stable MTS method for solving incompressible Navier-Stokes equations by coupling different IMEX-SAV schemes with different time steps using non-overlapping dual-Schur DD. Existing MTS methods that are primarily formulated for solving single-field problems, such as structural dynamics and transient heat conduction, cannot be directly applied to solve two-field (velocity-pressure) problems such as Navier-Stokes equations because additional constraints are needed to ensure the continuity of both solution fields across the interface between subdomains. Furthermore, existing MTS methods are designed for coupling explicit and/or implicit time integration schemes, but not for IMEX schemes because developing the necessary continuity constraints across subdomain interfaces that lead to stable and accurate coupling algorithms is a nontrivial task [41]. Even with general-purpose staggered coupling methods [58, 59], which are agnostic to the underlying time integration, stability and accuracy are usually problem-dependent and cannot always be guaranteed. Thus, we propose the MTS-IMEX-SAV method that combines the computational advantages of the MTS approach, avoids non-linear iterations using the IMEX scheme, and achieves unconditional stability with the SAV method. The proposed MTS-IMEX-SAV method advances the solution using large time steps for most of the problem domain whereas small time steps are used only for subdomains that need a fine temporal resolution. First, we describe the IMEX-BDF k -SAV scheme [26] for incompressible Navier-Stokes equations in Section 2 and present the fully discrete algebraic equations obtained using FEM. In Section 3, we formulate a novel MTS method for coupling IMEX schemes based on dual-Schur DD and SAV stabilization. Note that the proposed MTS-IMEX-SAV method leads to linear algebraic systems, thereby avoiding the need for iterations for non-linear terms in the Navier-Stokes equations. We prove that the proposed method is unconditionally stable and show convergence results to illustrate its accuracy. Finally, Section 4 presents several numerical examples to illustrate the performance of the proposed MTS-IMEX-SAV method in terms of stability, accuracy, and computational cost.

2. UTS method for IMEX-BDF k -SAV schemes with finite element discretizations

In this section, we briefly review basic equations of the IMEX-BDF k -SAV scheme [26, 31] with finite element discretizations for simulating incompressible Navier-Stokes equations, using a single time step for entire problem domain - called uniform time-step (UTS) method. For a given body force $\mathbf{b}$, the incompressible Navier-Stokes equations are

$$\frac{\partial \mathbf{u}}{\partial t} - \nu \nabla^2 \mathbf{u} + \mathbf{u} \cdot \nabla \mathbf{u} + \nabla p = \mathbf{b} \quad \text{in } \Omega \times (0, T] \quad (1a)$$

$$\nabla \cdot \mathbf{u} = 0 \quad \text{in } \Omega \times (0, T] \quad (1b)$$

128 with the boundary conditions

$$\mathbf{u} = \mathbf{u}_D \quad \text{on } \Gamma_D \quad (2a)$$

$$\boldsymbol{\sigma} \mathbf{n} = \boldsymbol{\tau}_N \quad \text{on } \Gamma_N \quad (2b)$$

129 and the initial condition

$$\mathbf{u}(t = 0) = \mathbf{u}_0 \quad (3)$$

130 where $\mathbf{u}$ and p are velocity field and normalized pressure fields ($p = P/\rho$, where P is the pressure and ρ is
 131 the density), respectively. The material parameter ν is the kinetic viscosity, $\boldsymbol{\sigma}$ is the viscous stress tensor
 132 ($= 2\nu\boldsymbol{\epsilon} - p\mathbf{I}$), $\boldsymbol{\epsilon}$ is the strain-rate tensor ($= \frac{1}{2}(\nabla\mathbf{u} + \nabla\mathbf{u}^T)$), $\mathbf{n}$ is the outward normal to Γ , T is the time
 133 duration, and Ω and Γ are the problem domain and its boundary, respectively. The subscripts D and N
 134 denote Dirichlet and Neumann boundaries, respectively. Note that $\Gamma_D \cup \Gamma_N = \Gamma$ and $\Gamma_N \cap \Gamma_D = \emptyset$.

135 2.1. IMEX-SAV method for the BDFk schemes

136 Let the energy $E[\mathbf{u}]$ be $\frac{1}{2} \int_{\Omega} \mathbf{u} \cdot \mathbf{u} d\Omega$. Then, the energy law is

$$\frac{dE[\mathbf{u}]}{dt} = \int_{\Omega} \frac{\partial \mathbf{u}}{\partial t} \cdot \mathbf{u} d\Omega = \int_{\Omega} (\nu \nabla^2 \mathbf{u} + \mathbf{b} - \mathbf{u} \cdot \nabla \mathbf{u} - \nabla p) \cdot \mathbf{u} d\Omega \quad (4)$$

137 Using the integral by parts and the Gauss theorem with the continuity equation, Eq. (4) can be rewritten as

$$\frac{dE[\mathbf{u}]}{dt} = - \int_{\Omega} \nu \nabla \mathbf{u} : \nabla \mathbf{u} d\Omega + \int_{\Omega} \mathbf{b} \cdot \mathbf{u} d\Omega - \int_{\Gamma_D} p \mathbf{u} \cdot \mathbf{n} d\Gamma_D + \int_{\Gamma_N} \boldsymbol{\tau}_N \cdot \mathbf{u} d\Gamma_N - \frac{1}{2} \int_{\Gamma} (\mathbf{u} \cdot \mathbf{u}) \mathbf{u} \cdot \mathbf{n} d\Gamma \quad (5)$$

138 where the operator $\mathbf{A} : \mathbf{B}$ denotes the inner product of tensors $\mathbf{A}$ and $\mathbf{B}$. Next, we define the scalar auxiliary
 139 variable as $R(t) = E[\mathbf{u}(t)] + C_0$ with $C_0 > 0$. Note that a positive constant C_0 shifts the energy of the
 140 system. Although the SAV approach leads to unconditional stability for any choice of $C_0 > 0$, in practice, a
 141 sufficiently large value of C_0 is chosen to achieve high accuracy [20, 41]. Since C_0 is problem-dependent, its
 142 effect on accuracy needs to be determined through a few exploratory simulations for specific applications.
 143 The evolution equation for $R(t)$ can be expressed as

$$\frac{dR(t)}{dt} = - \frac{R(t)}{E[\mathbf{u}(t)] + C_0} \mathcal{K}[\mathbf{u}(t)] \quad (6)$$

144 where

$$\mathcal{K}[\mathbf{u}] = - \frac{dE[\mathbf{u}]}{dt} = \int_{\Omega} (\nu \nabla \mathbf{u} : \nabla \mathbf{u} - \mathbf{b} \cdot \mathbf{u}) d\Omega + \int_{\Gamma_D} p \mathbf{u} \cdot \mathbf{n} d\Gamma_D - \int_{\Gamma_N} \boldsymbol{\tau}_N \cdot \mathbf{u} d\Gamma_N + \frac{1}{2} \int_{\Gamma} (\mathbf{u} \cdot \mathbf{u}) \mathbf{u} \cdot \mathbf{n} d\Gamma \quad (7)$$

145 In the SAV's evolution equation (6), the fractional term is introduced for numerical discretization, because
 146 the discrete SAV, R , is not always equal to $E + C_0$.

Given the SAV and its evolution equation, using the IMEX-BDF k -SAV schemes, the Navier-Stokes equations at $(n+1)^{\text{th}}$ step can be approximated as

$$\frac{\alpha \bar{\mathbf{u}}_{n+1} - \mathcal{A}[\bar{\mathbf{u}}_n]}{\Delta t} - \nu \nabla^2 \bar{\mathbf{u}}_{n+1} + \nabla \bar{p}_{n+1} = \mathbf{b}_{n+1} - \mathcal{B}[\mathbf{u}_n] \cdot \nabla \mathcal{B}[\mathbf{u}_n] \quad (8a)$$

$$\nabla \cdot \bar{\mathbf{u}}_{n+1} = 0 \quad (8b)$$

$$\frac{R_{n+1} - R_n}{\Delta t} = -\frac{R_{n+1}}{E[\bar{\mathbf{u}}_{n+1}] + C_0} \mathcal{K}[\bar{\mathbf{u}}_{n+1}] \quad (8c)$$

$$\xi_{n+1} = \frac{R_{n+1}}{E[\bar{\mathbf{u}}_{n+1}] + C_0}, \quad \eta_{n+1} = 1 - (1 - \xi_{n+1})^{k+1} \quad (8d)$$

$$\mathbf{u}_{n+1} = \eta_{n+1} \bar{\mathbf{u}}_{n+1}, \quad \mathbf{p}^{n+1} = \eta_{n+1} \bar{p}_{n+1} \quad (8e)$$

where α is the k^{th} harmonic number ($= \sum_{i=1}^k 1/i$), and operators $\mathcal{A}$ and $\mathcal{B}$ are used for the implicit BDF k scheme and explicit k^{th} -order extrapolation, respectively:

$$\mathcal{A}[\bar{\mathbf{u}}_n] = \sum_{i=0}^{k-1} (-1)^i \frac{1}{i+1} \binom{k}{i+1} \bar{\mathbf{u}}_{n-i}, \quad \mathcal{B}[\mathbf{u}_n] = \sum_{i=0}^{k-1} (-1)^i \binom{k}{i+1} \mathbf{u}_{n-i} \quad (9)$$

and binomial coefficient $\binom{k}{i+1}$ is $\frac{k!}{(k-i-1)!(i+1)!}$.

2.2. Finite element discretization

A weak form of the incompressible Navier-Stokes equations can be established using the Galerkin procedure - the linear momentum equations and continuity equation are weighted with the velocities and pressure, respectively [60]. We present the following finite element formulation for two-dimensional (2D) planar flows for simplicity and consistency with the numerical examples presented in the later sections. Nevertheless, the equations can readily be formulated for a general d -dimensional case, if needed. The discrete finite element equations are

$$\begin{bmatrix} \hat{\mathbf{K}} & \mathbf{H} \\ \mathbf{H}^T & \mathbf{0} \end{bmatrix} \begin{bmatrix} \bar{\mathbf{u}}_{n+1} \\ \bar{p}_{n+1} \end{bmatrix} = \begin{bmatrix} \hat{\mathbf{f}}_{n+1} \\ \mathbf{0} \end{bmatrix} + \frac{1}{\Delta t} \begin{bmatrix} \mathbf{M} \\ \mathbf{0} \end{bmatrix} \mathcal{A}[\bar{\mathbf{u}}_n] \quad (10a)$$

$$\frac{R_{n+1} - R_n}{\Delta t} = -\frac{R_{n+1}}{E[\bar{\mathbf{u}}_{n+1}] + C_0} \mathcal{K}[\bar{\mathbf{u}}_{n+1}, \bar{p}_{n+1}] \quad (10b)$$

$$\xi_{n+1} = \frac{R_{n+1}}{E[\bar{\mathbf{u}}_{n+1}] + C_0}, \quad \eta_{n+1} = 1 - (1 - \xi_{n+1})^{k+1} \quad (10c)$$

$$\mathbf{u}_{n+1} = \eta_{n+1} \bar{\mathbf{u}}_{n+1}, \quad \mathbf{p}_{n+1} = \eta_{n+1} \bar{p}_{n+1} \quad (10d)$$

where $\mathbf{u}$ and $\mathbf{p}$ denote the nodal velocity and pressure vector, respectively,

$$\hat{\mathbf{K}} = \frac{\alpha}{\Delta t} \mathbf{M} + \mathbf{K}, \quad \hat{\mathbf{f}}_{n+1} = \mathbf{f}_{n+1} - \mathbf{C} [\mathcal{B}[\mathbf{u}_n]] \mathcal{B}[\mathbf{u}_n] \quad (11a)$$

$$\mathbf{M} = \begin{bmatrix} \mathbf{M}_u & \mathbf{0} \\ \mathbf{0} & \mathbf{M}_p \end{bmatrix}, \quad \mathbf{K} = \begin{bmatrix} \mathbf{K}_u & \mathbf{0} \\ \mathbf{0} & \mathbf{K}_p \end{bmatrix}, \quad \mathbf{H} = \begin{bmatrix} \mathbf{H}_x \\ \mathbf{H}_y \end{bmatrix}, \quad \mathbf{u} = \begin{bmatrix} \mathbf{u}_x \\ \mathbf{u}_y \end{bmatrix} \quad (11b)$$

and $\mathbf{u}_x$ and $\mathbf{u}_y$ denote the nodal velocity vectors in x - and y -directions, respectively. In Eqs. (11a) and (11b), $\mathbf{M}_u$, $\mathbf{K}_u$, $\mathbf{H}_x$, $\mathbf{H}_y$, $\mathbf{C}$, and $\mathbf{f}$ are the global matrices and vectors, which can be computed by the

summation process using local matrices and vectors. Let the corresponding local matrices and vectors of element $[e]$ be $\mathbf{M}_u^{[e]}$, $\mathbf{K}_u^{[e]}$, $\mathbf{H}_x^{[e]}$, $\mathbf{H}_y^{[e]}$, $\mathbf{C}^{[e]}$, and $\mathbf{f}^{[e]}$:

$$\mathbf{M}_u^{[e]} = \int_{\Omega^{[e]}} \mathbf{N}_u^{[e]T} \mathbf{N}_u^{[e]} d\Omega, \quad \mathbf{K}_u^{[e]} = \int_{\Omega^{[e]}} \nu^{[e]} \left(\frac{\partial \mathbf{N}_u^{[e]T}}{\partial x} \frac{\partial \mathbf{N}_u^{[e]}}{\partial x} + \frac{\partial \mathbf{N}_u^{[e]T}}{\partial y} \frac{\partial \mathbf{N}_u^{[e]}}{\partial y} \right) d\Omega \quad (12a)$$

$$\mathbf{H}_x^{[e]} = - \int_{\Omega^{[e]}} \frac{\partial \mathbf{N}_u^{[e]T}}{\partial x} \mathbf{N}_p^{[e]} d\Omega, \quad \mathbf{H}_y^{[e]} = - \int_{\Omega^{[e]}} \frac{\partial \mathbf{N}_u^{[e]T}}{\partial y} \mathbf{N}_p^{[e]} d\Omega \quad (12b)$$

$$\mathbf{C}^{[e]} [\mathbf{u}^{[e]}] = \int_{\Omega^{[e]}} \left(\mathbf{N}_u^{[e]T} \mathbf{N}_u^{[e]} \mathbf{u}_x^{[e]} \frac{\partial \mathbf{N}_u^{[e]T}}{\partial x} + \mathbf{N}_u^{[e]T} \mathbf{N}_u^{[e]} \mathbf{u}_y^{[e]} \frac{\partial \mathbf{N}_u^{[e]T}}{\partial y} \right) d\Omega \quad (12c)$$

$$\mathbf{f}^{[e]} = \int_{\Omega^{[e]}} \mathbf{N}_u^{[e]T} \mathbf{b}^{[e]} d\Omega + \int_{\Gamma_N^{[e]}} \mathbf{N}_{Su}^{[e]T} \boldsymbol{\tau}_N^{[e]} d\Gamma \quad (12d)$$

where $\nu^{[e]}$ is the kinetic viscosity. Shape functions $\mathbf{N}_u^{[e]}$, $\mathbf{N}_{Su}^{[e]}$, and $\mathbf{N}_p^{[e]}$ are for velocity, surface velocity, and pressure interpolation matrices, respectively, in element $[e]$.

In Eq. (10b), the energy E and its dissipation $\mathcal{K}$ with finite element discretizations can be represented by, respectively,

$$E[\bar{\mathbf{u}}] = \frac{1}{2} \bar{\mathbf{u}}^T \mathbf{M} \bar{\mathbf{u}} \quad (13a)$$

$$\begin{aligned} \mathcal{K}[\bar{\mathbf{u}}, \bar{\mathbf{p}}] = & \bar{\mathbf{u}}^T (\mathbf{K} \bar{\mathbf{u}} - \mathbf{f}) + \sum_{[e]} \int_{\Gamma_D^{[e]}} \mathbf{N}_{Sp}^{[e]} \bar{\mathbf{p}}^{[e]} \mathbf{N}_{Su}^{[e]} \left(\bar{\mathbf{u}}_x^{[e]} n_x^{[e]} + \bar{\mathbf{u}}_y^{[e]} n_y^{[e]} \right) d\Gamma \\ & + \sum_{[e]} \int_{\Gamma^{[e]}} \frac{(\mathbf{N}_{Su}^{[e]} \bar{\mathbf{u}}_x^{[e]})^2 + (\mathbf{N}_{Su}^{[e]} \bar{\mathbf{u}}_y^{[e]})^2}{2} \mathbf{N}_{Su}^{[e]} \left(\bar{\mathbf{u}}_x^{[e]} n_x^{[e]} + \bar{\mathbf{u}}_y^{[e]} n_y^{[e]} \right) d\Gamma \end{aligned} \quad (13b)$$

where $\mathbf{N}_{Sp}^{[e]}$, $n_x^{[e]}$ and $n_y^{[e]}$ are, respectively, the surface pressure interpolation matrix, x - and y -component of the normal vector on surface $\Gamma^{[e]}$ in element $[e]$. In this study, to compute global matrices and vectors, we use Taylor-Hood finite elements for discretizing the governing equations in space, to satisfy inf-sup condition [61].

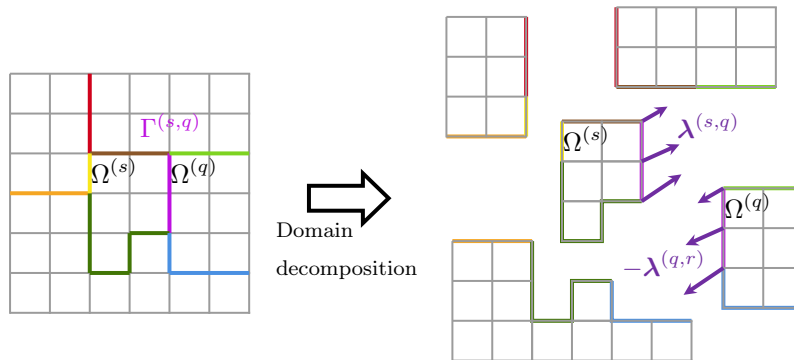


Fig. 1 A general partitioned problem domain.

3. MTS method for the incompressible Navier-Stokes equations

In this section, we develop a new unconditionally stable MTS method to couple multiple IMEX schemes using a non-overlapping dual-Schur domain decomposition (DD) and SAV stabilization for the incompressible Navier-Stokes equations. In contrast to existing approaches discussed in the Introduction, such as LTS and MRT, the proposed MTS method is the first instance of work that enables the use of IMEX schemes with different time steps in different regions of the spatial problem domain. The key novelty is in the derivation of the necessary interface equations between subdomains that ensure continuity of the solution and achieve unconditional stability with the SAV approach.

As seen in Fig. 1, a problem domain is decomposed into several spatial subdomains with Lagrange multipliers λ on interfaces, where different IMEX time stepping methods and/or different time steps are used. After decomposing the problem domain Ω into several subdomains $\Omega^{(1)}, \Omega^{(2)}, \dots, \Omega^{(S)}$, the semi-discrete governing equations for a subdomain $\Omega^{(s)}$ (where $s = 1, 2, \dots, S$) are

$$\begin{aligned} \mathbf{M}^{(s)} \frac{\partial \mathbf{u}^{(s)}(t)}{\partial t} + \mathbf{K}^{(s)} \mathbf{u}^{(s)}(t) + \mathbf{C}^{(s)} [\mathbf{u}^{(s)}(t)] \mathbf{u}^{(s)}(t) + \mathbf{H}^{(s)} \mathbf{p}^{(s)}(t) &= \mathbf{f}^{(s)}(t) - \sum_{q=1, q \neq s}^S \mathbf{B}_u^{(s,q)T} \lambda^{u(s,q)}(t) \\ \mathbf{H}^{(s)T} \mathbf{u}^{(s)}(t) &= - \sum_{q=1, q \neq s}^S \mathbf{B}_p^{(s,q)T} \lambda^{p(s,q)}(t) \end{aligned} \quad (14)$$

where superscripts $[\cdot]^{(s)}$ and $[\cdot]^{(s,q)}$ denote subdomain $\Omega^{(s)}$ and interface $\Gamma^{(s,q)}$. Matrices $\mathbf{B}_u^{(s,q)}$ and $\mathbf{B}_p^{(s,q)}$ are the connectivity matrices across the interface $\Gamma^{(s,q)}$ for velocity and pressure, respectively. Vectors $\lambda^{u(s,q)}$ and $\lambda^{p(s,q)}$ are the Lagrange multipliers in velocity and pressure, respectively. Note that $\lambda^{u(s,q)} = \lambda^{u(q,s)}$ and $\lambda^{p(s,q)} = \lambda^{p(q,s)}$.

On the interface $\Gamma^{(s,q)}$ between arbitrary two subdomains $\Omega^{(s)}$ and $\Omega^{(q)}$, continuity of velocity and pressure is imposed to glue both subdomains:

$$\mathbf{B}_u^{(s,q)} \bar{\mathbf{u}}^{(s)}(t) + \mathbf{B}_u^{(q,s)} \bar{\mathbf{u}}^{(q)}(t) = \mathbf{0} \quad \text{for } s = 1, 2, \dots, S \text{ and } q = 1, 2, \dots, S, \text{ with } q \neq s \quad (15a)$$

$$\mathbf{B}_p^{(s,q)} \bar{\mathbf{p}}^{(s)}(t) + \mathbf{B}_p^{(q,s)} \bar{\mathbf{p}}^{(q)}(t) = \mathbf{0} \quad (15b)$$

In this study, a finite element discretization is used resulting in Boolean connectivity matrices $\mathbf{B}_u^{(s,q)}$ and $\mathbf{B}_p^{(s,q)}$ [35]. Note, however, that Eqs. (14) and (15) are not specific to the finite element formulation and are also obtained from other spatial discretizations such as meshfree methods [39, 62].

3.1. New stable MTS-IMEX-SAV method for the BDFk schemes

In this section, we describe the temporal discretization and solution algorithm for the proposed MTS-IMEX-SAV method. For temporal discretization of Eq. (14), individual subdomains are integrated in time with different time stepping schemes and/or different time-step sizes, as shown in Fig. 2. Without loss of generality, we assume $\Delta t^{(1)} \geq \Delta t^{(s)}$ for any value of s . In the proposed MTS method, solutions are advanced from t_n to $t_{n+1} = t_n + \Delta t^{(1)} = t_n + m^{(2)} \Delta t^{(2)} = \dots = t_n + m^{(S)} \Delta t^{(S)}$, where $m^{(s)} = \Delta t^{(1)} / \Delta t^{(s)}$ is a

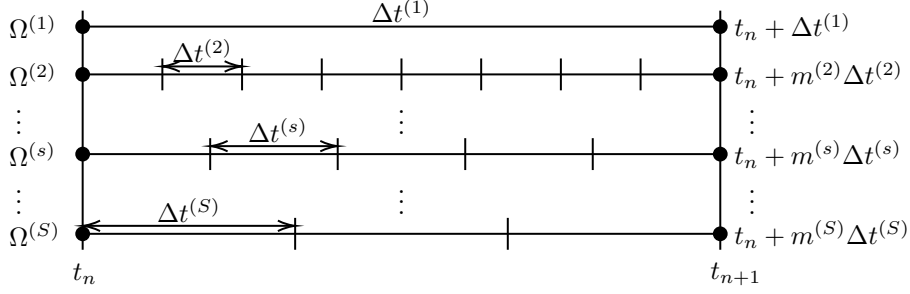


Fig. 2 Representation of multi-time-steps for S subdomains. Note that $m^{(s)} = \Delta t^{(1)} / \Delta t^{(s)}$.

time-step ratio for subdomain $\Omega^{(s)}$. Note that intermediate solutions for subdomain $\Omega^{(s)}$ are advanced from $t_{n+(j-1)/m}^{(s)} = t_n + (j-1)\Delta t^{(s)}$ to $t_{n+j/m}^{(s)} = t_n + j\Delta t^{(s)}$, where $j = 1, 2, \dots, m^{(s)}$. At time instance $t_{n+j/m}^{(s)}$, an quantity $[\cdot]$ is denoted as $[\cdot]_{n+j/m}^{(s)}$ (e.g. velocity $\mathbf{u}_{n+j/m}^{(s)}$, pressure $\mathbf{p}_{n+j/m}^{(s)}$, external force $\mathbf{f}_{n+j/m}^{(s)}$, and Lagrange multiplier $\boldsymbol{\lambda}_{n+j/m}^{(s,q)}$).

Using IMEX-BDF k time integration schemes, the fully discrete governing equations for subdomain $\Omega^{(s)}$ at time $t_{n+j/m}^{(s)}$ ($j = 1, 2, \dots, m^{(s)}$) are

$$\begin{bmatrix} \hat{\mathbf{K}}^{(s)} & \mathbf{H}^{(s)} \\ \mathbf{H}^{(s)^T} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \bar{\mathbf{u}}_{n+j/m}^{(s)} \\ \bar{\mathbf{p}}_{n+j/m}^{(s)} \end{bmatrix} = \begin{bmatrix} \hat{\mathbf{f}}_{n+j/m}^{(s)} \\ \mathbf{0} \end{bmatrix} + \frac{1}{\Delta t^{(s)}} \begin{bmatrix} \mathbf{M}^{(s)} \\ \mathbf{0} \end{bmatrix} \mathcal{A}[\bar{\mathbf{u}}_{n+(j-1)/m}^{(s)}] - \sum_{q=1, q \neq s}^S \begin{bmatrix} \mathbf{B}_u^{(s,q)^T} \boldsymbol{\lambda}_{n+j/m}^{u(s,q)} \\ \mathbf{B}_p^{(s,q)^T} \boldsymbol{\lambda}_{n+j/m}^{p(s,q)} \end{bmatrix} \quad (16)$$

where

$$\hat{\mathbf{K}}^{(s)} = \frac{\alpha^{(s)}}{\Delta t^{(s)}} \mathbf{M}^{(s)} + \mathbf{K}^{(s)}, \quad \hat{\mathbf{f}}_{n+j/m}^{(s)} = \mathbf{f}_{n+j/m}^{(s)} - \mathbf{C}^{(s)} \left[\mathcal{B}^{(s)}[\tilde{\mathbf{u}}_{n+(j-1)/m}^{(s)}] \right] \mathcal{B}^{(s)}[\tilde{\mathbf{u}}_{n+(j-1)/m}^{(s)}] \quad (17)$$

with the following implicit operator $\mathcal{A}^{(s)}$ and explicit operator $\mathcal{B}^{(s)}$:

$$\mathcal{A}^{(s)}[\bar{\mathbf{u}}_{n+(j-1)/m}^{(s)}] = \sum_{i=1}^{k^{(s)}} (-1)^{i-1} \frac{1}{i} \binom{k^{(s)}}{i} \bar{\mathbf{u}}_{n+(j-i)/m}^{(s)}, \quad \mathcal{B}^{(s)}[\tilde{\mathbf{u}}_{n+(j-1)/m}^{(s)}] = \sum_{i=1}^{k^{(s)}} (-1)^{i-1} \binom{k^{(s)}}{i} \tilde{\mathbf{u}}_{n+(j-i)/m}^{(s)} \quad (18)$$

where $k^{(s)}$ is the order of the BDF scheme in subdomain $\Omega^{(s)}$ and $\tilde{\mathbf{u}}^{(s)}$ is an explicit predictor. While in principle any explicit predictor may be used, we found that certain choices of predictors lead to better solutions as shown later.

In the proposed MTS method, Lagrange multipliers $\boldsymbol{\lambda}^u$ and $\boldsymbol{\lambda}^p$ on interface $\Gamma^{(s,q)}$ in subdomain $\Omega^{(s)}$ at intermediate time instant $t_{n+j/m}^{(s)}$ are computed using a linear interpolation between t_n and t_{n+1} is

$$\boldsymbol{\lambda}_{n+j/m}^{u(s,q)} = \frac{m^{(s)} - j}{m^{(s)}} \boldsymbol{\lambda}_n^{u(s,q)} + \frac{j}{m^{(s)}} \boldsymbol{\lambda}_{n+1}^{u(s,q)}, \quad \boldsymbol{\lambda}_{n+j/m}^{p(s,q)} = \frac{m^{(s)} - j}{m^{(s)}} \boldsymbol{\lambda}_n^{p(s,q)} + \frac{j}{m^{(s)}} \boldsymbol{\lambda}_{n+1}^{p(s,q)} \quad (19)$$

The use of linear interpolation preserves the convergence rates of the first-order and second-order IMEX schemes, as shown in Section 3.4. Note that linear interpolation of the Lagrange multipliers does not necessarily imply that global accuracy is limited to second-order. For example, a MTS method for the third-order time integration scheme using linear interpolation of Lagrange multipliers gives third-order accurate

solutions [40, 41]. Substituting Eq. (19) into Eq. (16) leads to the following fully discrete equation in $\Omega^{(s)}$:

$$\mathbb{K}^{(s)} \bar{\mathbf{x}}_{n+j/m}^{(s)} = \underbrace{\mathfrak{G}_{n+j/m}^{(s)}}_{\text{Decoupled}} + \underbrace{\mathbb{M}^{(s)} \mathcal{A}^{(s)} [\mathbb{I}^{(s)} \bar{\mathbf{x}}_{n+(j-1)/m}^{(s)}]}_{\text{Coupled}} - \frac{j}{m^{(s)}} \sum_{q=1, q \neq s}^S \mathbb{B}^{(s,q)T} \mathbb{L}_{n+1}^{(s,q)} \quad \text{for } j = 1, 2, \dots, m^{(s)} \quad (20)$$

where

$$\mathbb{K}^{(s)} = \begin{bmatrix} \hat{\mathbf{K}}^{(s)} & \mathbf{H}^{(s)} \\ \mathbf{H}^{(s)T} & \mathbf{0} \end{bmatrix}, \quad \bar{\mathbf{x}}_{n+j/m}^{(s)} = \begin{bmatrix} \hat{\mathbf{u}}_{n+j/m}^{(s)} \\ \hat{\mathbf{p}}_{n+j/m}^{(s)} \end{bmatrix}, \quad (21a)$$

$$\mathfrak{G}_{n+j/m}^{(s)} = \hat{\mathbf{f}}_{n+j/m}^{(s)} - \frac{m^{(s)} - j}{m^{(s)}} \sum_{q=1, q \neq s}^S \mathbb{B}^{(s,q)T} \mathbb{L}_n^{(s,q)}, \quad \hat{\mathbf{f}}_{n+j/m}^{(s)} = \begin{bmatrix} \hat{\mathbf{f}}_{n+j/m}^{(s)} \\ \mathbf{0} \end{bmatrix}, \quad (21b)$$

$$\mathbb{M}^{(s)} = \frac{1}{\Delta t^{(s)}} \begin{bmatrix} \mathbf{M}^{(s)} \\ \mathbf{0} \end{bmatrix}, \quad \mathbb{I}^{(s)} = \begin{bmatrix} \mathbf{I}^{(s)} & \mathbf{0} \end{bmatrix}, \quad \mathbb{B}^{(s,q)} = \begin{bmatrix} \mathbf{B}_u^{(s,q)} & \mathbf{0} \\ \mathbf{0} & \mathbf{B}_p^{(s,q)} \end{bmatrix}, \quad \mathbb{L}_{n+1}^{(s,q)} = \begin{bmatrix} \lambda_{n+1}^{u(s,q)} \\ \lambda_{n+1}^{p(s,q)} \end{bmatrix} \quad (21c)$$

and $\mathbf{I}^{(s)}$ is the identity matrix of size of the number of nodal velocity in subdomain $\Omega^{(s)}$. Note that in Eq. (20), the first term on the right-hand side is denoted as ‘decoupled’ because it depends only on known external forces $\hat{\mathbf{f}}_{n+j/m}^{(s)}$ and known Lagrange multipliers from the previous time step $\mathbb{L}_n^{(s,q)}$. These are both associated only with subdomain $\Omega^{(s)}$ and thus the ‘decoupled’ term can be computed independently from other neighboring subdomains $\Omega^{(q)}$. On the other hand, the ‘coupled’ term requires computation of Lagrange multipliers between subdomains $\Omega^{(s)}$ and $\Omega^{(q)}$ at the next time step $\mathbb{L}_{n+1}^{(s,q)}$, which must be solved simultaneously considering the coupling between subdomains as follows.

To compute Lagrange multipliers $\mathbb{L}_{n+1}^{(s,q)}$, constraint conditions (15) are enforced across interface $\Gamma^{(s,q)}$ at every largest time step $\Delta t^{(1)}$:

$$\mathbb{B}^{(s,q)} \bar{\mathbf{x}}_{n+1}^{(s)} + \mathbb{B}^{(q,s)} \bar{\mathbf{x}}_{n+1}^{(q)} = \mathbf{0} \quad (22)$$

Therefore, Eqs. (20) and (22) for all subdomains and interfaces lead to a linear system of equation in the form of a saddle-point problem. Solving this system, one computes coupled solutions $\bar{\mathbf{x}}_{n+j/m}^{(s)} = [\hat{\mathbf{u}}_{n+j/m}^{(s)}; \hat{\mathbf{p}}_{n+j/m}^{(s)}]$ for $j = 1, 2, \dots, m^{(s)}$ and $s = 1, 2, \dots, S$.

To solve Eqs. (20) and (22), we use the following predictor-corrector algorithm (here, predictor and corrector are corresponding to decoupled and coupled parts and do not refer to classical predictor-corrector projection methods). First, we decompose the solution $\bar{\mathbf{x}}_{n+j/m}^{(s)}$ into decoupled solution $\bar{\bar{\mathbf{x}}}_{n+j/m}^{(s)}$ and coupled solution $\bar{\mathbf{x}}_{n+j/m}^{\prime(s)}$:

$$\bar{\mathbf{x}}_{n+j/m}^{(s)} = \bar{\bar{\mathbf{x}}}_{n+j/m}^{(s)} + \bar{\mathbf{x}}_{n+j/m}^{\prime(s)} \quad (23)$$

where decoupled solution $\bar{\bar{\mathbf{x}}}_{n+j/m}^{(s)}$ are obtained by solving the following equations:

$$\mathbb{K}^{(s)} \bar{\bar{\mathbf{x}}}_{n+j/m}^{(s)} = \mathbb{M}^{(s)} \mathcal{A}^{(s)} [\mathbb{I}^{(s)} \bar{\bar{\mathbf{x}}}_{n+(j-1)/m}^{(s)}] + \mathfrak{G}_{n+j/m}^{(s)} \quad \text{for } j = 1, 2, \dots, m^{(s)} \quad (24)$$

with $\bar{\bar{\mathbf{x}}}_{n+\ell}^{(s,q)} = \bar{\mathbf{x}}_{n+\ell}^{(s,q)}$ for $\ell \leq 0$. For the effect of the Lagrange multipliers, coupled solution $\bar{\mathbf{x}}_{n+j/m}^{\prime(s)}$ is defined

as

$$\bar{\mathbf{x}}_{n+j/m}^{\prime(s)} = - \sum_{q=1, q \neq s}^S \mathbb{Y}_{j/m}^{(s,q)} \mathbb{L}_{n+1}^{(s,q)} \quad (25)$$

where block matrices $\mathbb{Y}_{j/m}^{(s,q)}$ on interface $\Gamma^{(s,q)}$ are computed by solving the following equations to incorporate the effect of Lagrange multipliers:

$$\mathbb{K}^{(s)} \mathbb{Y}_{j/m}^{(s,q)} = \mathbb{M}^{(s)} \mathcal{A}^{(s)} [\mathbb{I}^{(s)} \mathbb{Y}_{(j-1)/m}^{(s,q)}] + \frac{j}{m^{(s)}} \mathbb{B}^{(s,q)^T} \quad \text{for } j = 1, 2, \dots, m^{(s)} \quad (26)$$

with $\mathbb{Y}_\ell^{(s,q)} = \mathbf{0}$ for $\ell \leq 0$. Note that Eq. (26) is solved only once. Then, substitution of Eq. (23) into Eq. (22) leads to

$$\mathbb{B}^{(s,q)} \sum_{r=1, r \neq s}^S \mathbb{Y}_1^{(s,r)} \mathbb{L}_{n+1}^{(s,r)} + \mathbb{B}^{(q,s)} \sum_{r=1, r \neq q}^S \mathbb{Y}_1^{(q,r)} \mathbb{L}_{n+1}^{(q,r)} = \mathbb{B}^{(s,q)} \bar{\bar{\mathbb{X}}}_{n+1}^{(s)} + \mathbb{B}^{(q,s)} \bar{\bar{\mathbb{X}}}_{n+1}^{(q)} \quad (27)$$

for all interfaces. Since $\mathbb{L}^{(s,q)} = \mathbb{L}^{(q,s)}$, Eq. (27) for all interfaces leads to an interface problem in the following linear system of equations:

$$\mathbb{F}^I \mathbb{L}_{n+1}^I = \mathbb{h}_{n+1}^I \quad (28)$$

where

$$\mathbb{L}_{n+1}^I = \begin{bmatrix} \mathbb{L}_{n+1}^{(1,2)} \\ \mathbb{L}_{n+1}^{(1,3)} \\ \vdots \\ \mathbb{L}_{n+1}^{(1,S)} \\ \vdots \\ \mathbb{L}_{n+1}^{(s,s+1)} \\ \mathbb{L}_{n+1}^{(s,s+2)} \\ \vdots \\ \mathbb{L}_{n+1}^{(s,S)} \\ \vdots \\ \mathbb{L}_{n+1}^{(S-1,S)} \end{bmatrix}, \quad \mathbb{h}_{n+1}^I = \begin{bmatrix} \mathbb{B}^{(1,2)} \bar{\bar{\mathbb{X}}}_{n+1}^{(1)} + \mathbb{B}^{(2,1)} \bar{\bar{\mathbb{X}}}_{n+1}^{(2)} \\ \mathbb{B}^{(1,3)} \bar{\bar{\mathbb{X}}}_{n+1}^{(1)} + \mathbb{B}^{(3,1)} \bar{\bar{\mathbb{X}}}_{n+1}^{(3)} \\ \vdots \\ \mathbb{B}^{(1,S)} \bar{\bar{\mathbb{X}}}_{n+1}^{(1)} + \mathbb{B}^{(S,1)} \bar{\bar{\mathbb{X}}}_{n+1}^{(S)} \\ \vdots \\ \mathbb{B}^{(s,s+1)} \bar{\bar{\mathbb{X}}}_{n+1}^{(s)} + \mathbb{B}^{(s+1,s)} \bar{\bar{\mathbb{X}}}_{n+1}^{(s+1)} \\ \mathbb{B}^{(s,s+2)} \bar{\bar{\mathbb{X}}}_{n+1}^{(s)} + \mathbb{B}^{(s+2,s)} \bar{\bar{\mathbb{X}}}_{n+1}^{(s+2)} \\ \vdots \\ \mathbb{B}^{(s,S)} \bar{\bar{\mathbb{X}}}_{n+1}^{(s)} + \mathbb{B}^{(S,s)} \bar{\bar{\mathbb{X}}}_{n+1}^{(S)} \\ \vdots \\ \mathbb{B}^{(S-1,S)} \bar{\bar{\mathbb{X}}}_{n+1}^{(S-1)} + \mathbb{B}^{(S,S-1)} \bar{\bar{\mathbb{X}}}_{n+1}^{(S)} \end{bmatrix} \quad (29)$$

and $\mathbb{F}^I = \mathbb{B}^I \mathbb{Y}^I$ (see Appendix A for block matrices $\mathbb{B}^I$ and $\mathbb{Y}^I$). Note that factorization of $\mathbb{F}^I$ needs to be carried out only once.

Finally, for stabilizing solutions $\bar{\bar{\mathbb{X}}}_{n+j/m}^{(s)}$ in Eq. (23), we implement the SAV approach for the MTS method. First, for the entire problem domain, we define a scalar auxiliary variable $R = E + C_0$, where $E = \sum_{s=1}^S E^{(s)}$. Its evolution equation (6) is discretized using backward difference scheme to advance the SAV R from t_n to t_{n+1} :

$$\frac{R_{n+1} - R_n}{\Delta t^{(1)}} = - \frac{R_{n+1}}{E[\bar{\mathbf{u}}_{n+1}] + C_0} \mathcal{K}[\bar{\mathbf{u}}_{n+1}, \bar{\mathbf{p}}_{n+1}] \quad (30)$$

where $E[\bar{\mathbf{u}}_{n+1}] = \sum_{s=1}^S E^{(s)}[\bar{\mathbf{u}}_{n+1}^{(s)}]$ and $\mathcal{K}[\bar{\mathbf{u}}_{n+1}, \bar{\mathbf{p}}_{n+1}] = \sum_{s=1}^S \mathcal{K}^{(s)}[\bar{\mathbf{u}}_{n+1}^{(s)}, \bar{\mathbf{p}}_{n+1}^{(s)}]$. Then, we compute final

251 solution $\mathbf{x}_{n+j/m}^{(s)}$ in subdomain $\Omega^{(s)}$ at time $t_{n+j/m}^{(s)}$ from the following steps:

$$\xi_{n+1} = \frac{R^{n+1}}{E[\bar{\mathbf{u}}_{n+1}] + C_0} \quad (31a)$$

$$\xi_{n+j/m}^{(s)} = \frac{m^{(s)} - j}{m^{(s)}} \xi_n + \frac{j}{m^{(s)}} \xi_{n+1} \quad \text{for } j = 1, 2, \dots, m^{(s)} \quad (31b)$$

$$\eta_{n+j/m}^{(s)} = 1 - (1 - \xi_{n+j/m}^{(s)})^{k^{(s)}+1} \quad (31c)$$

$$\mathbf{x}_{n+j/m}^{(s)} = \eta_{n+j/m}^{(s)} \bar{\mathbf{x}}_{n+j/m}^{(s)} \quad (31d)$$

252 Note that for individual subdomain $\Omega^{(s)}$, variable $\xi_{n+j/m}^{(s)}$ is obtained from the linear interpolation between
 253 ξ_n and ξ_{n+1} . The proposed SAV approach, using the linear interpolation for $\xi_{n+j/m}^{(s)}$, preserves the local
 254 truncation error of the MTS method (see Section 3.3).

255 Returning to the definition of an explicit predictor in Eq. (17) that is used to treat the non-linear terms
 256 for the MTS method, we introduce a new predictor $\tilde{\mathbf{u}}_{n+\ell}^{(s)}$ as an input of explicit operator $\mathcal{B}$:

$$\tilde{\mathbf{u}}_{n+\ell}^{(s)} = \begin{cases} \mathbf{u}_{n+\ell}^{(s)} & \text{for } \ell \leq 0 \\ \eta_n^{(s)} \left(\bar{\mathbf{u}}_{n+\ell}^{(s)} + \bar{\mathbf{u}}'_{n-1+\ell} \right) & \text{for } \ell > 0 \end{cases} \quad \text{for } \ell = (j-1)/m, (j-2)/m, \dots, (j-k^{(s)})/m \quad (32)$$

257 where $\bar{\mathbf{u}}_{n+\ell}^{(s)}$ is the current decoupled solution (see Eq. (24)) and $\bar{\mathbf{u}}'_{n-1+\ell}^{(s)}$ is the previous coupled solution (see
 258 Eq. (25)). Scalar factor $\eta_n^{(s)}$ is considered as the effect of the SAV stabilization at the previous time t_n (see
 259 Eq.(31c)).

260 From Eqs. (17) and (21b), the explicit operator $\mathcal{B}^{(s)}[\tilde{\mathbf{u}}_{n+(j-1)/m}^{(s)}]$, which is an input to convection matrix
 261 $\mathbf{C}^{(s)}$, is used to compute $\bar{\mathbf{u}}_{n+j/m}^{(s)}$ for $j = 1, 2, \dots, m^{(s)}$ in the predictor stage (24). In the MTS algorithm, the
 262 solution is computed in the following order:

$$\underbrace{\bar{\mathbf{u}}_{n+1/m}^{(s)} \rightarrow \bar{\mathbf{u}}_{n+2/m}^{(s)} \rightarrow \dots \rightarrow \bar{\mathbf{u}}_{n+1}^{(s)}}_{\text{Predictor stage}} \rightarrow \underbrace{\bar{\mathbf{u}}'_{n+j/m}^{(s)} \forall j}_{\text{Corrector stage}} \rightarrow \underbrace{\bar{\mathbf{u}}_{n+j/m}^{(s)} \forall j}_{\text{SAV stage}} \rightarrow \underbrace{\eta_{n+j/m}^{(s)} \forall j}_{\text{Update stage}} \rightarrow \underbrace{\mathbf{u}_{n+j/m}^{(s)} \forall j}_{\text{Update stage}} \quad (33)$$

263 Thus, to ensure an explicit treatment of the non-linear term, the input to the $\mathcal{B}^{(s)}$ at time instant $t_{n+j/m}$
 264 should be independent of $\bar{\mathbf{u}}^{(s)}$ at $t \geq t_{n+j/m}$, as well as of $\bar{\mathbf{u}}'^{(s)}$, $\bar{\mathbf{u}}^{(s)}$, $\eta^{(s)}$, and $\mathbf{u}^{(s)}$ at $t > t_n$. When a
 265 time instant is less than or equal to current time instant t_n (i.e. $\ell \leq 0$), there is no need for predicting
 266 state vectors (i.e. $\tilde{\mathbf{u}}_{n+\ell}^{(s)} = \mathbf{u}_{n+\ell}^{(s)}$). On the other hand, when $\ell > 0$, we predict the state vectors as $\tilde{\mathbf{u}}_{n+\ell}^{(s)} =$
 267 $\eta_n^{(s)} \left(\bar{\mathbf{u}}_{n+\ell}^{(s)} + \bar{\mathbf{u}}'_{n-1+\ell} \right)$ to avoid using non-linear iteration algorithm. It is important to note that instead
 268 of $\tilde{\mathbf{u}}_{n+\ell}^{(s)}$, using $\mathbf{u}_{n+\ell}^{(s)} (= \eta_{n+\ell}^{(s)} \left(\bar{\mathbf{u}}_{n+\ell}^{(s)} + \bar{\mathbf{u}}'_{n+\ell} \right))$ for non-linear terms leads to the need for non-linear iteration
 269 algorithmic because $\mathcal{B}^{(s)}[\mathbf{u}_{n+(j-1)/m}^{(s)}]$ includes intermediate solutions $\mathbf{u}_{n+1/m}^{(s)}$, $\mathbf{u}_{n+2/m}^{(s)}$, ..., or $\mathbf{u}_{n+(m-1)/m}^{(s)}$ unless
 270 $m^{(s)} = 1$ for any subdomain $\Omega^{(s)}$. The solution is updated at each block-time-step cycle and this process is
 271 repeated for the next block-time-step $\Delta t^{(1)}$.

272 Fig. 3 summarizes the computational procedure for the proposed MTS-IMEX-SAV method. Note that
 273 since the quantities of $\mathbf{x}^{(s)}$ and $\mathbf{y}^{(s,q)}$ are independent of subdomains and interfaces, one can use parallel
 274 processing for independent quantities to accelerate simulations.

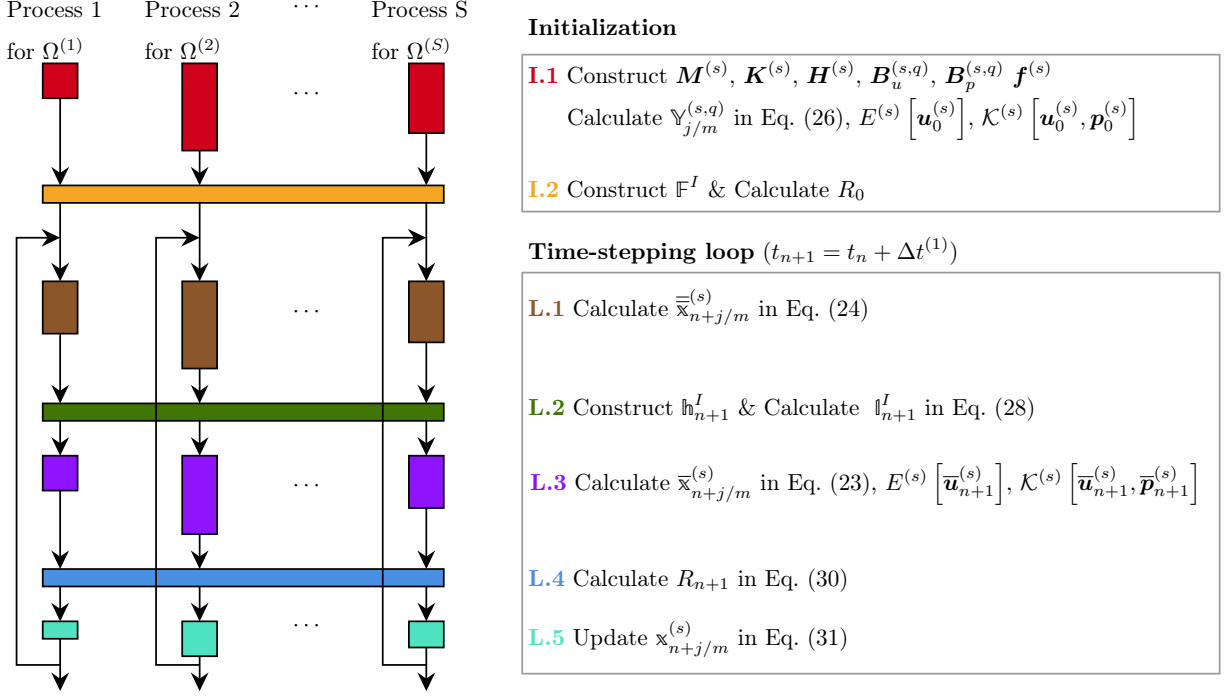


Fig. 3 Flowchart of the proposed MTS-IMEX-SAV method.

With the MTS method, subdomain partitioning may be optimized by taking into account both the number of degrees of freedom (DOF) and the time step in each subdomain. This approach ensures a well-balanced computational load across all processors. For instance, a recursive strategy that produces almost optimal MTS decompositions and leads to balanced loads on processors is described in [63]. The proposed MTS-IMEX-SAV method can also be applied together with dynamic DD where the subdomain partitioning evolves in time. In this case, the time-stepping loop would need to include the reconstruction of the following matrices: $\mathbf{M}^{(s)}, \mathbf{K}^{(s)}, \mathbf{H}^{(s)}, \mathbf{B}_u^{(s,q)}, \mathbf{B}_p^{(s,q)}, \mathbf{f}^{(s)}, \mathbb{V}_{j/m}^{(s,q)}$, and the assembly of $\mathbb{F}^I$. This process can be very computationally intensive for large DOF on the subdomain interfaces. An analysis of the computational cost associated with $\mathbb{V}_{j/m}^{(s,q)}$ is presented in [41]. For dynamic DD, an iterative approach to solve the dynamically evolving subdomain and interface operators may be more viable.

3.2. Proof of stability

In this section, we analyze stability characteristics considering no-slip boundary conditions for the velocity [27, 31, 64]. We have the following stability results of the proposed MTS-IMEX-SAV method in the absence and presence of the external force imposing no-slip boundary conditions.

Theorem 3.1. *In the absence of the external force $\mathbf{f}$ with no-slip boundary velocity $\mathbf{u}$, the MTS-IMEX-SAV method for any $k^{(s)}$ are unconditionally energy stable in the sense that*

$$R_{n+1} \leq R_n \quad (34)$$

291 Furthermore, there exists $U_E^{(s)} > 0$ for any subdomain $\Omega^{(s)}$ such that

$$E^{(s)}[\mathbf{u}_{n+1}^{(s)}] \leq \left(U_E^{(s)}\right)^2 \quad (35)$$

292 *Proof.* From Eq. (13), $E + C_0 > 0$ and $\mathcal{K} \geq 0$ at all times for no external force and no-slip boundary
293 conditions. Thus, given $R_n > 0$, Eq. (30) leads to

$$R_{n+1} = \frac{R_n}{1 + \Delta t^{(1)} \frac{\mathcal{K}[\bar{\mathbf{u}}_{n+1}, \bar{\mathbf{p}}_{n+1}]}{E[\bar{\mathbf{u}}_{n+1}] + C_0}} \quad (36)$$

294 ensuring $R_{n+1} > 0$. In addition, from Eq. (31a), $\xi_{n+1} > 0$. Thus, Eq. (30) leads to

$$R_{n+1} - R_n = -\Delta t^{(1)} \xi_{n+1} \mathcal{K}[\bar{\mathbf{u}}_{n+1}, \bar{\mathbf{p}}_{n+1}] \leq 0 \quad (37)$$

295 This proves Eq. (34).

296 From Eqs. (31a) and (34),

$$|\xi_{n+1}| = \frac{R_{n+1}}{E[\bar{\mathbf{u}}_{n+1}, \bar{\mathbf{p}}_{n+1}] + C_0} \leq \frac{R_0}{E[\bar{\mathbf{u}}_{n+1}] + C_0} \quad (38)$$

297 Moreover, $\eta_{n+1}^{(s)}$ can be represented as $\eta_{n+1}^{(s)} = \xi_{n+1} P^{(s)}(\xi_{n+1})$ from Eqs. (31b) and (31c), where $P^{(s)}$ is a
298 polynomial of degree $k^{(s)}$. Thus, there exist $U_E^{(s)}$ such that $|P^{(s)}(\xi_{n+1})| \leq 2\sqrt{C_0}U_E^{(s)}/R_0$ for all subdomains.
299 From Eq. (38), we derive the following inequality:

$$|\eta_{n+1}^{(s)}| = |\xi_{n+1} P^{(s)}(\xi_{n+1})| \leq \frac{2\sqrt{C_0}U_E^{(s)}}{E[\bar{\mathbf{u}}_{n+1}] + C_0} \leq \frac{2\sqrt{C_0}U_E^{(s)}}{E^{(s)}[\bar{\mathbf{u}}_{n+1}^{(s)}] + C_0} \quad (39)$$

300 Along with Eqs. (13a) and (31d), Eq. (39) implies

$$E^{(s)}[\mathbf{u}_{n+1}^{(s)}] = \left(\eta_{n+1}^{(s)}\right)^2 E^{(s)}[\bar{\mathbf{u}}_{n+1}^{(s)}] \leq \left(\frac{2\sqrt{C_0}U_E^{(s)}}{E^{(s)}[\bar{\mathbf{u}}_{n+1}^{(s)}] + C_0}\right)^2 E^{(s)}[\bar{\mathbf{u}}_{n+1}^{(s)}] \leq \left(U_E^{(s)}\right)^2 \quad (40)$$

301 The proof is complete. $\square$

302 **Theorem 3.2.** In the presence of the the external force $\mathbf{f}$ with no-slip boundary velocity $\mathbf{u}$, when $C_0 \geq$
303 $\max_{s=1}^S C_0^{(s)}$, where $C_0^{(s)} = 2(S\Delta t^{(1)})^2 \|\mathbf{L}^{(s)-1} \mathbf{f}^{(s)}\|^2 \forall t \in [0, T]$ and $\mathbf{M}^{(s)} = \mathbf{L}^{(s)} \mathbf{L}^{(s)T}$ (Cholesky decom-
304 position), there exists $U_E^{(s)} > 0$ for any subdomain $\Omega^{(s)}$ such that

$$E^{(s)}[\mathbf{u}_n^{(s)}] \leq \left(U_E^{(s)}\right)^2 \quad \forall n \in [0, T/\Delta t^{(1)}] \quad (41)$$

305 *Proof.* From Eq. (13a) with selection of $C_0 \geq \max_{s=1}^S C_0^{(s)}$, where $C_0^{(s)} = 2(S\Delta t^{(1)})^2 \|\mathbf{L}^{(s)-1} \mathbf{f}^{(s)}\|^2$, we
306 obtain

$$\begin{aligned} E[\bar{\mathbf{u}}_{n+1}] + C_0 &\geq E^{(s)}[\bar{\mathbf{u}}_{n+1}^{(s)}] + C_0^{(s)} \geq \frac{1}{2} \bar{\mathbf{u}}_{n+1}^{(s)T} \mathbf{M}^{(s)} \bar{\mathbf{u}}_{n+1}^{(s)} + 2(S\Delta t^{(1)})^2 \|\mathbf{L}^{(s)-1} \mathbf{f}^{(s)}\|^2 \\ &\geq \frac{1}{2} \|\mathbf{L}^{(s)T} \bar{\mathbf{u}}_{n+1}^{(s)}\|^2 + 2(S\Delta t^{(1)})^2 \|\mathbf{L}^{(s)-1} \mathbf{f}^{(s)}\|^2 \end{aligned} \quad (42)$$

307 In addition, Eq. (13b) with no-slip boundary conditions leads to

$$\mathcal{K}[\bar{\mathbf{u}}, \bar{\mathbf{p}}] = \bar{\mathbf{u}}^T (\mathbf{K} \bar{\mathbf{u}} - \mathbf{f}) \quad (43)$$

Since $\mathbf{K}^{(s)}$ is a positive semi-definite, we obtain

$$|\mathcal{K}^{(s)}[\bar{\mathbf{u}}^{(s)}, \bar{\mathbf{p}}^{(s)}]| \leq |\bar{\mathbf{u}}^{(s)T} \mathbf{f}^{(s)}| = |\bar{\mathbf{u}}^{(s)T} \mathbf{L}^{(s)} \mathbf{L}^{(s)-1} \mathbf{f}^{(s)}| \leq \|\mathbf{L}^{(s)T} \bar{\mathbf{u}}^{(s)}\| \|\mathbf{L}^{(s)-1} \mathbf{f}^{(s)}\| \quad (44)$$

Thus, from Eqs. (42) and (44), we obtain

$$\left| \frac{\mathcal{K}^{(s)}[\bar{\mathbf{u}}_{n+1}^{(s)}, \bar{\mathbf{p}}_{n+1}^{(s)}]}{E[\bar{\mathbf{u}}_{n+1}] + C_0} \right| \leq \frac{\|\mathbf{L}^{(s)T}(\bar{\mathbf{u}}_{n+1}^{(s)})\| \|\mathbf{L}^{(s)-1} \mathbf{f}_{n+1}^{(s)}\|}{\frac{1}{2} \|\mathbf{L}^{(s)T} \bar{\mathbf{u}}_{n+1}^{(s)}\|^2 + 2(S\Delta t^{(1)})^2 \|\mathbf{L}^{(s)-1} \mathbf{f}_{n+1}^{(s)}\|^2} \leq \frac{1}{2S\Delta t^{(1)}} \quad (45)$$

which implies that

$$\left| \frac{\mathcal{K}[\bar{\mathbf{u}}_{n+1}, \bar{\mathbf{p}}_{n+1}]}{E[\bar{\mathbf{u}}_{n+1}] + C_0} \right| \leq \sum_{s=1}^S \left| \frac{\mathcal{K}^{(s)}[\bar{\mathbf{u}}_{n+1}^{(s)}, \bar{\mathbf{p}}_{n+1}^{(s)}]}{E[\bar{\mathbf{u}}_{n+1}] + C_0} \right| \leq \frac{1}{2\Delta t^{(1)}} \quad (46)$$

Given $R_n > 0$ with Eq. (46), we obtain from Eq. (36) with Eq. (45) that

$$R_{n+1} = R_n \left(1 + \Delta t^{(1)} \frac{\mathcal{K}[\bar{\mathbf{u}}_{n+1}, \bar{\mathbf{p}}_{n+1}]}{E[\bar{\mathbf{u}}_{n+1}] + C_0} \right)^{-1} > 0 \quad (47)$$

Moreover, the summation of Eq. (30) from $n = 0$ to $n = m$ leads to

$$R_{m+1} = R_0 - \Delta t^{(1)} \sum_{i=0}^m \frac{\mathcal{K}[\bar{\mathbf{u}}_{i+1}, \bar{\mathbf{p}}_{i+1}]}{E[\bar{\mathbf{u}}_{i+1}] + C_0} R_{i+1} \leq R_0 + \Delta t^{(1)} \sum_{i=0}^m \left| \frac{\mathcal{K}[\bar{\mathbf{u}}_{i+1}, \bar{\mathbf{p}}_{i+1}]}{E[\bar{\mathbf{u}}_{i+1}] + C_0} \right| R_{i+1} \leq R_0 + \frac{1}{2} \sum_{i=0}^m R_{i+1} \quad (48)$$

Applying the discrete Gronwall lemma (see [65]), one proves

$$R_m \leq 2^m R_0 \quad \forall m \in [0, T/\Delta t^{(1)}] \quad (49)$$

Thus,

$$|\xi_{n+1}| = \frac{R_{n+1}}{E[\bar{\mathbf{u}}_{n+1}] + C_0} \leq \frac{2^{n+1} R_0}{E[\bar{\mathbf{u}}_{n+1}] + C_0} \quad (50)$$

Recall $\eta_{n+1}^{(s)} = \xi_{n+1} P^{(s)}(\xi_{n+1})$. From Eq. (50), we derive that there exists $U_E^{(s)} > 0$ such that

$$|\eta_{n+1}^{(s)}| = |\xi_{n+1} P^{(s)}(\xi_{n+1})| \leq \frac{2\sqrt{C_0} U_E^{(s)}}{E[\bar{\mathbf{u}}_{n+1}] + C_0} \quad (51)$$

Along with Eqs. (13a) and (31d), Eq. (51) implies

$$E^{(s)}[\mathbf{u}_{n+1}^{(s)}] = \left(\eta_{n+1}^{(s)} \right)^2 E^{(s)}[\bar{\mathbf{u}}_{n+1}^{(s)}] \leq \left(\frac{2\sqrt{C_0} U_E^{(s)}}{E^{(s)}[\bar{\mathbf{u}}_{n+1}^{(s)}] + C_0} \right)^2 E^{(s)}[\bar{\mathbf{u}}_{n+1}^{(s)}] \leq \left(U_E^{(s)} \right)^2 \quad (52)$$

The proof is complete. $\square$

In the proposed MTS-IMEX-SAV method, numerical stability is achieved without any CFL restriction. Since the SAV approach guarantees stabilization of the numerical solution over each block-time-step $\Delta t^{(1)}$ (i.e. $\dots t_{n-1}, t_n, t_{n+1}, \dots$), Theorems 3.1 and 3.2 remains valid, regardless of the selections of time integration scheme, the interpolation of Lagrange multipliers $\lambda_{n+j/m}^{u(s,q)}$ and $\lambda_{n+j/m}^{p(s,q)}$, the interpolation of $\xi_{n+j/m}^{(s)}$, and the predictor $\tilde{\mathbf{u}}^{(s)}$.

3.3. Local truncation error

In this section, the local truncation error (LTE) of the proposed MTS-IMEX-SAV method is analyzed. We prove that the SAV method preserve the accuracy order of the underlying MTS method. Let the underlying MTS method be k th-order accurate:

$$\begin{aligned}\bar{\mathbf{u}}_{n+j/m}^{(s)} - \mathbf{u}^{(s)}(t_{n+j/m}^{(s)}) &\propto C_u \Delta t^{k^{(s)}+1} + O(\Delta t^{k^{(s)}+2}) \\ \bar{\mathbf{p}}_{n+j/m}^{(s)} - \mathbf{p}^{(s)}(t_{n+j/m}^{(s)}) &\propto C_p \Delta t^{k^{(s)}+1} + O(\Delta t^{k^{(s)}+2})\end{aligned}\quad (53)$$

where C_u and C_p are constant coefficients of the leading-order terms in the LTEs in velocity and pressure, respectively. Substituting Eq. (36) into Eq. (31a), one obtains

$$\xi_{n+1} = \frac{R_n}{E[\bar{\mathbf{u}}_{n+1}] + C_0 + \Delta t^{(1)} \mathcal{K}[\bar{\mathbf{u}}_{n+1}, \bar{\mathbf{p}}_{n+1}]} \propto \frac{R_n}{E[\bar{\mathbf{u}}_{n+1}] + C_0 + O(\Delta t^{(1)})} \quad (54)$$

Using Eq. (13b), the above equation can be rewritten as

$$\xi_{n+1} \propto \frac{\mathbf{u}_n^T \mathbf{M} \mathbf{u}_n}{\bar{\mathbf{u}}_{n+1}^T \mathbf{M} \bar{\mathbf{u}}_{n+1} + 2C_0 + O(\Delta t^{(1)})} \quad (55)$$

Substituting of Eq. (53) into Eq.(55), one obtains

$$\xi_{n+1} \propto \frac{\mathbf{u}_n^T \mathbf{M} \mathbf{u}_n}{\mathbf{u}(t_{n+1})^T \mathbf{M} \mathbf{u}(t_{n+1}) + 2C_0 + O(\Delta t^{(1)})} \quad (56)$$

The exact solution can be obtained from a Taylor series expansion about t_n as follows:

$$\mathbf{u}_{n+1} = \mathbf{u}_n + \sum_{k=1}^{\infty} \frac{(\Delta t^{(1)})^k}{k!} \frac{d^k \mathbf{u}(t_n)}{dt^k} \quad (57)$$

Thus, the following terms in denominator of Eq. (56) can be rewritten as

$$\mathbf{u}(t_{n+1})^T \mathbf{M} \mathbf{u}(t_{n+1}) \propto \mathbf{u}_n^T \mathbf{M} \mathbf{u}_n + O(\Delta t^{(s)}) \quad (58)$$

Thus, Eq. (56) can be simplified to

$$\xi_{n+1} \propto \frac{\mathbf{u}_n^T \mathbf{M} \mathbf{u}_n}{\mathbf{u}_n^T \mathbf{M} \mathbf{u}_n + 2C_0 + C_E \Delta t^{(1)} + O((\Delta t^{(1)})^2)} \propto 1 - \frac{C_E}{2R_n} \Delta t^{(1)} + O((\Delta t^{(1)})^2) \propto 1 + O(\Delta t^{(1)}) \quad (59)$$

where $R_n = \frac{1}{2} \mathbf{u}_n^T \mathbf{M} \mathbf{u}_n + C_0$ and C_E is the coefficient of the leading-term. Since $\xi_n = 1$, the following equation is obtained from Eq. (31b):

$$\xi_{n+j/m}^{(s)} \propto 1 + O(\Delta t^{(1)}) \quad (60)$$

Since $\eta_{n+j/m}^{(s)} = 1 - (1 - \xi_{n+j/m}^{(s)})^{k^{(s)}+1}$, Eq. (60) implies:

$$\eta_{n+j/m}^{(s)} \propto 1 + O((\Delta t^{(1)})^{k^{(s)}+1}) \propto 1 + C_\eta (\Delta t^{(1)})^{k^{(s)}+1} + O((\Delta t^{(1)})^{k^{(s)}+2}) \quad (61)$$

where C_η is the coefficient of the leading-term.

Finally, from Eqs. (53) and (61), LTEs in velocity and pressure are obtained as:

$$\begin{aligned}LTE_u &= \mathbf{u}_{n+j/m}^{(s)} - \mathbf{u}^{(s)}(t_{n+j/m}^{(s)}) = \eta_{n+j/m}^{(s)} \bar{\mathbf{u}}_{n+j/m}^{(s)} - \mathbf{u}^{(s)}(t_{n+j/m}^{(s)}) \propto (C_u + C_\eta \bar{\mathbf{u}}_{n+j/m}^{(s)}) \Delta t^{k^{(s)}+1} + O(\Delta t^{k^{(s)}+2}) \\ LTE_p &= \mathbf{p}_{n+j/m}^{(s)} - \mathbf{p}^{(s)}(t_{n+j/m}^{(s)}) = \eta_{n+j/m}^{(s)} \bar{\mathbf{p}}_{n+j/m}^{(s)} - \mathbf{p}^{(s)}(t_{n+j/m}^{(s)}) \propto (C_p + C_\eta \bar{\mathbf{p}}_{n+j/m}^{(s)}) \Delta t^{k^{(s)}+1} + O(\Delta t^{k^{(s)}+2})\end{aligned}\quad (62)$$

Therefore, the proposed MTS-IMEX-SAV method preserves the LTEs of the underlying MTS method.

3.4. Accuracy tests

In this section, we investigate convergence rates of the proposed MTS-IMEX-SAV method for BDF1 and BDF2 schemes using numerical tests rather than a rigorous proof. The following manufactured analytical solution is considered in $\Omega = [-1, 1]^2$ with $\nu = 0.01$ from $t = 0$ to $t = 1$:

$$\begin{aligned} u_x(t) &= \cos^2(\pi x/2) \sin(\pi y) \sin(t)/2 \\ u_y(t) &= -\sin(\pi x) \cos^2(\pi y/2) \sin(t)/2 \\ p(t) &= \cos(\pi x) \sin(\pi y) \sin(t)/2 \end{aligned} \tag{63}$$

where u_x and u_y denote x - and y -components of $\mathbf{u}$, respectively. The body force $\mathbf{b}$ is selected such that the incompressible Navier-Stokes equations in Eq. (1) are satisfied. We set $C_0 = 10^4$ for the SAV stabilization, and for spatial discretizations, we employ two types structured meshes with the following time-step ratios: 1) P2-P1 elements with two subdomains ($\Omega^{(1)} = [-1, 0] \times [-1, 1]$ and $\Omega^{(2)} = [0, 1] \times [-1, 1]$) and $m^{(2)} = 2$; and 2) Q2-Q1 elements with four subdomains ($\Omega^{(1)} = [-1, 0] \times [-1, 0]$, $\Omega^{(2)} = [0, 1] \times [-1, 0]$, $\Omega^{(3)} = [-1, 0] \times [0, 1]$, and $\Omega^{(4)} = [0, 1] \times [0, 1]$) and $m^{(2)} = 2$, $m^{(3)} = 5$, and $m^{(4)} = 10$. In this section, subdomain partitioning has not been optimized to yield balanced computational loads and is presented only as a test case for accuracy. For the spatial discretization, Q2-Q1 elements are used in the uniform square mesh, in which the length of the side of elements is 0.0125, and the P2-P1 elements are used in the triangular mesh, which is obtained by partitioning the problem domain into squares and subsequently drawing a diagonal in each square.

To quantify errors, we define the following two measures of global instantaneous error in the solution:

$$L^2(x) = \|x_N - x(t_N)\|_2, \quad L^\infty(x) = \|x_N - x(t_N)\|_\infty \tag{64}$$

where x is the quantity of interest (e.g. velocity $\mathbf{u}$ or pressure p) and $x(t)$ is the analytical solution, and N is the number of $\Delta t^{(1)}$ time steps over the duration of the simulation.

Using the P2-P1 elements with 2 subdomains, convergence rates of the proposed MTS method are summarized in Table 1. We notice that the convergence rate of the MTS methods is the minimum value of the convergence rates of the underlying time integration schemes. Thus, hereafter, we use the same-order BDF time integration schemes for every subdomain. Fig. 4 shows the convergence rates of the MTS-IMEX-SAV methods using P2-P1 and Q2-Q1 elements. The methods based on BDF1 and BDF2 schemes achieve first- and second-order accuracy, respectively. Note that in Fig. 4(b) and (d), the convergence curve shows a deflection because for small $\Delta t^{(1)}$, errors from spatial discretization dominate over those from temporal discretization.

Table 1 Convergence rates of the proposed MTS-IMEX-SAV methods (rounded to the nearest integer) for the BDF1 and BDF2 schemes with two subdomains using the P2-P1 elements.

Time integration		L^2 error				L^∞ error			
$\Omega^{(1)}$	$\Omega^{(2)}$	$\mathbf{u}^{(1)}$	$p^{(1)}$	$\mathbf{u}^{(2)}$	$p^{(2)}$	$\mathbf{u}^{(1)}$	$p^{(1)}$	$\mathbf{u}^{(2)}$	$p^{(2)}$
BDF1	BDF1	1	1	1	1	1	1	1	1
	BDF2	1	1	1	1	1	1	1	1
BDF2	BDF1	1	1	1	1	1	1	1	1
	BDF2	2	2	2	2	2	2	2	2

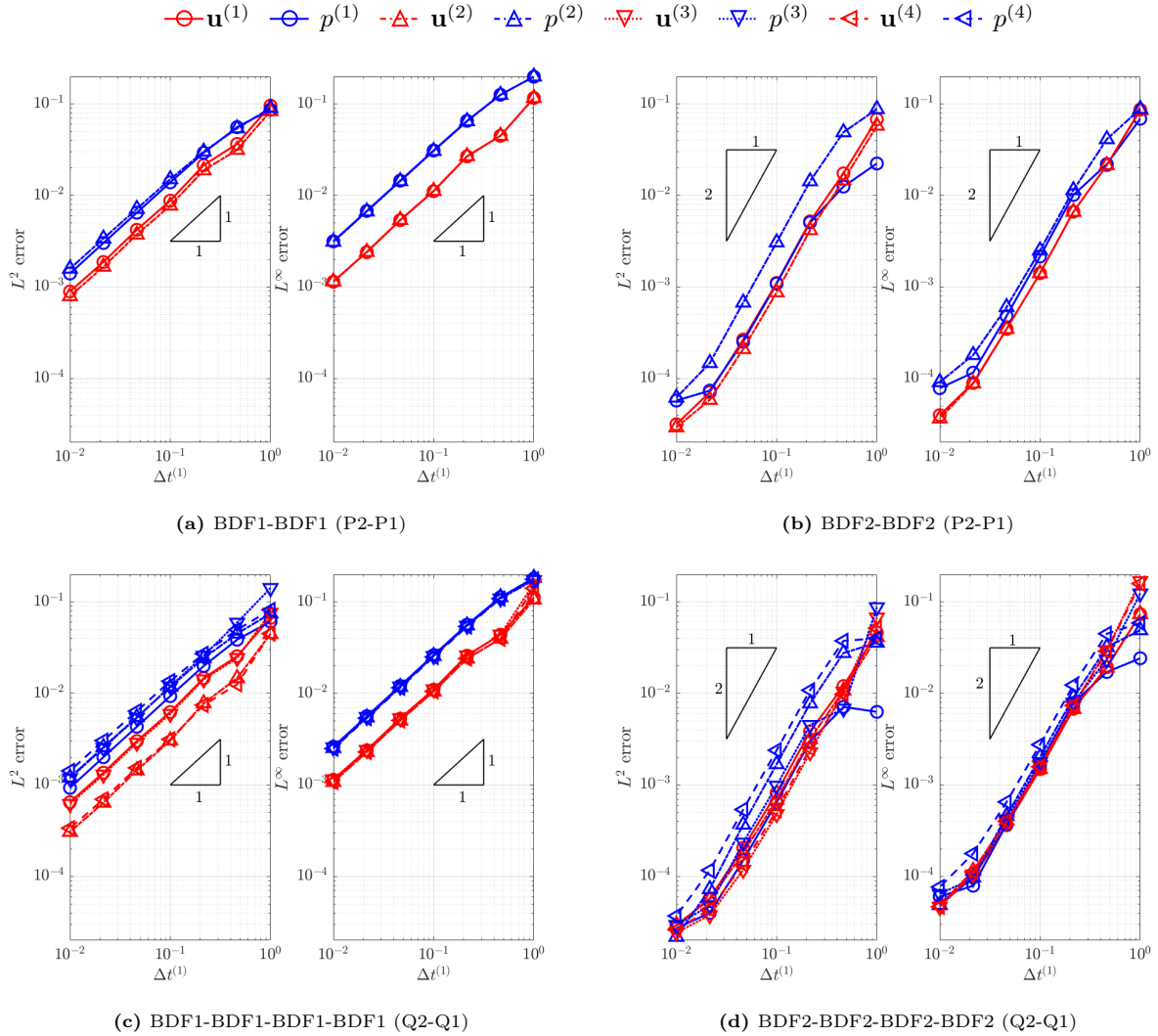


Fig. 4 L^2 and L^∞ errors in velocity and pressure of the proposed MTS-IMEX-SAV methods.

4. Representative numerical examples

In this section, we present four representative numerical examples to demonstrate the performance of the proposed MTS-IMEX-SAV method for BDF schemes with $C_0 = 10^4$ for the SAV stabilization. In this paper, a conservative value of $C_0 = 10^4$ is found to be sufficient. In general, however, the value of C_0 is problem dependent and requires some exploratory studies, similar to [20, 27, 41, 64]. While iterative solvers are typically preferred for large problems, in this study, we use direct solvers for all computational stages (i.e. Eqs. (24), (26), and (28)). We simulate all example problems on a Linux desktop with AMD EPYC 7702P with 64 cores, 128 threads, and 256 GB RAM. For the UTS method with domain decomposition and the MTS method, parallel computing are used by creating a local parallel pool. The total number of CPU threads N_T is evenly divided among the workers corresponding to the subdomains to maximize parallel efficiency. Thus, the number of threads assigned to each subdomain is $N_s = \lfloor N_T/S \rfloor$, where $\lfloor (\cdot) \rfloor$ is the the greatest integer less than or equal to $(\cdot)$. For a fair comparison, for the UTS method without domain decomposition, we use the same total number of CPU threads as in the UTS with domain decomposition and MTS methods (i.e. $N_s \times S$).

4.1. Stokes problem

We consider an infinity long oscillating plate in an infinite domain of fluid [66, 67]. When the plate is located at $y = 0$ and its velocity is $U \cos \omega t$ in the x -direction, where ω is the angular frequency, the exact steady-state solution is

$$u_x(y, t) = U e^{-\sqrt{\frac{\omega}{2\nu}} y} \cos \left(\omega t - \sqrt{\frac{\omega}{2\nu}} y \right) \quad (65)$$

In this section, we assume that $U = 1$, $\omega = 2\pi$, and $\nu = \pi$, and the computational domain is $\Omega = [-5, 5] \times [0, 10]$. For boundary conditions, we impose the no-slip boundary condition at the bottom: $u_x(0, t) = \cos t$. Since the problem domain is finite, the boundary condition at the top is $u_x(10, t) = U e^{-10} \cos(t - 10)$, which is derived from the exact solution at $y = 10$, instead of $u_x(\infty, t) = 0$. To die the transient solution, we impose an initial condition as $u_x(y, 0) = U e^{-y} \cos(-y)$.

As seen in Fig. 5a, for spatial discretization, we use 400 Q2-Q1 elements with a bias factor of 1.1 in the y -direction. For domain decomposition, we divide the problem domain into three subdomains. For the UTS method, time-step size is chosen as $\Delta t = \text{CFL} \frac{h}{U}$, where h denotes the minimum value of the characteristics lengths of elements. The characteristic length of a Q2-Q1 element is defined as $h = 0.5 A_{el} / \max_{i=1}^4 L_i$, where A_{el} and L_i are the area and the length of i th side of the element, respectively. Note that $h = h^{(3)}$ because $h^{(3)} < h^{(2)} < h^{(1)}$. On the other hand, for the MTS method, we select different time steps for different subdomains: $\Delta t^{(3)} = \Delta t$ with $m^{(2)} = 2$ and $m^{(3)} = 3$ because $h^{(1)}/h^{(2)} \approx 2$ and $h^{(1)}/h^{(3)} \approx 3$.

Fig. 5b shows the numerical solutions for the conventional IMEX-SAV method [26], IMEX-SAV method with domain decomposition (DD), and proposed MTS-IMEX-SAV method for the BDF2 schemes with CFL = 0.5. We observe that all methods closely match the exact solution. In addition, Fig. 6 shows L^2 and L^∞ errors at $t \approx 1.571$ and computational costs of the methods for several CFL numbers. The convergence

399 rates of all methods for the BDF1 and BDF2 schemes are observed to be 1 and 2. When the CFL number
 400 is the same, errors of the MTS method are larger than the UTS methods. However, as shown in Fig. 6, the
 401 proposed MTS method requires a low computational cost with a similar accuracy, since the time-step size
 402 of each cycle for the MTS method is three times that for the UTS method (i.e. $\Delta t^{(1)} = 3\Delta t$).

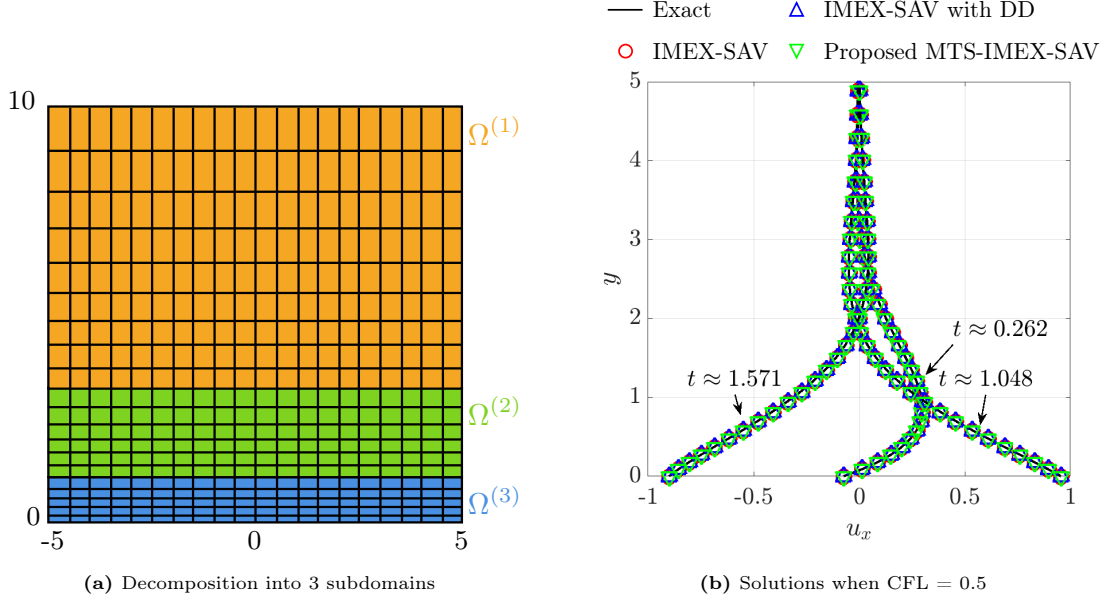


Fig. 5 Finite element models and exact and numerical solutions at $t \approx 0.262, 1.048$, and 1.571 in the Stokes problem.

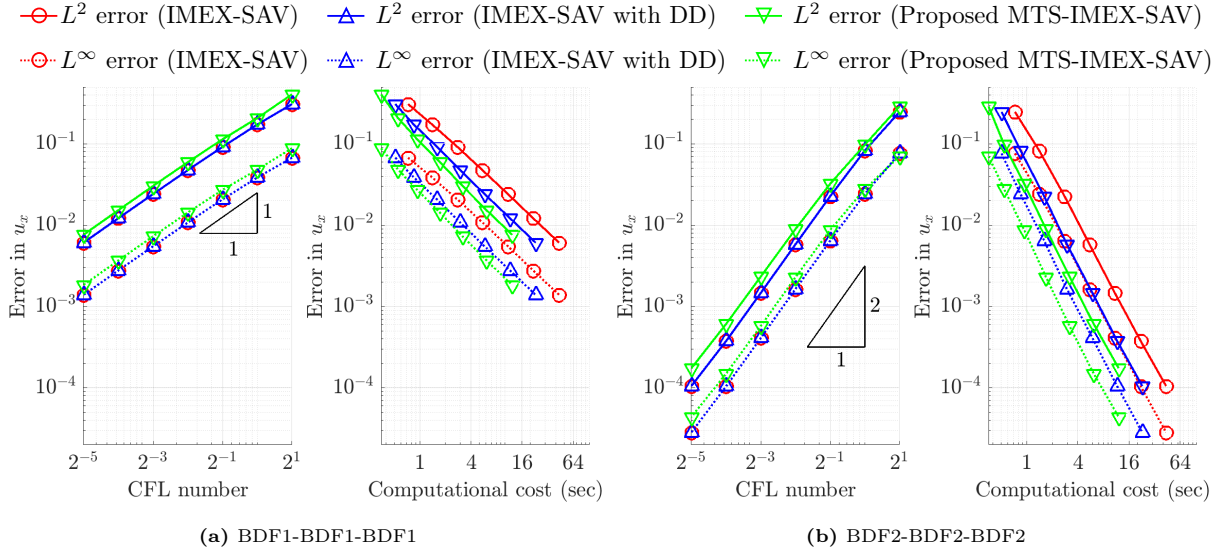


Fig. 6 L^2 and L^∞ errors in u_x at $t \approx 1.571$ in the Stokes problem.

4.2. Lid-driven cavity problem

As seen in Fig. 7a, we solve the lid-driven cavity flow, which is a benchmark problem for steady incompressible Navier-Stokes equations. In this section, we compare numerical results using the conventional IMEX-SAV method, IMEX-SAV method with DD, and proposed MTS-IMEX-SAV method for the BDF2 scheme with results provided by Ghia et al. in [68] at $Re = 100, 400, \text{ and } 1,000$, where Re denotes the Reynolds number ($= LU/\nu$). We consider triangular meshes with $70 \times 70 \times 2$ P2-P1 elements for spatial discretizations. A bias factor of 1.05 is used for all edges to make elements finer near four boundaries (see Fig. 7b). For domain decomposition, as shown in Fig. 8, we divide the entire domain into 16 subdomains.

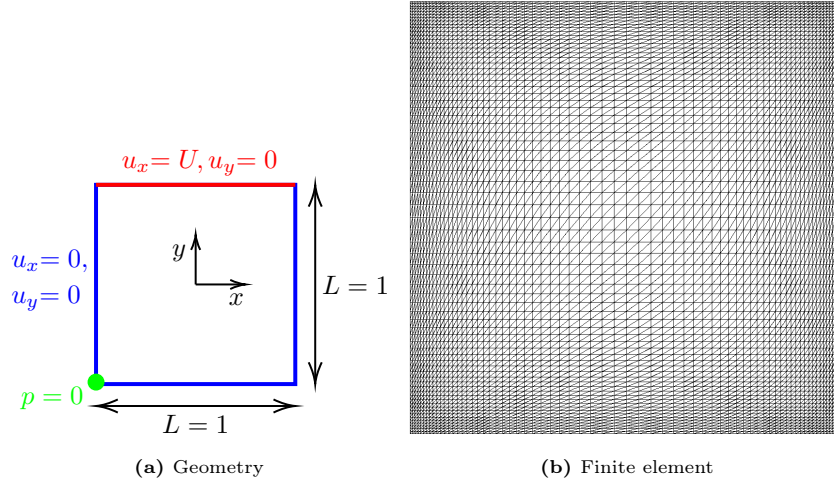


Fig. 7 Geometry and finite element model for the lid-driven cavity flow. The top wall moves at a velocity of $U = 1$.

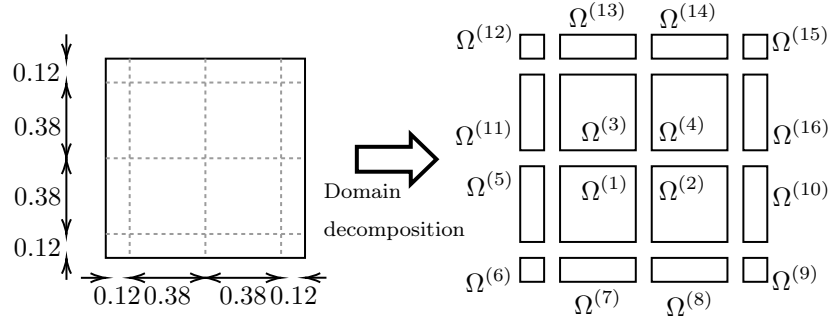


Fig. 8 Domain decomposition into 12 subdomains for the lid-driven cavity flow.

For the UTS methods, the time step is selected such that the CFL number is 2: $\Delta t = CFL \times h/U$ where $U = 1$. Note that the characteristic length of a P2-P1 element is defined as $h = A_{el}/\max_{i=1}^3 L_i$. On the other hand, the proposed MTS method, we choose $\Delta t^{(s)} = \Delta t$ for $s = 5, 6, \dots, 12$ with $m^{(1)} = m^{(2)} = m^{(3)} = m^{(4)} = 1$ and $m^{(5)} = m^{(6)} = \dots = m^{(12)} = 2$ because $h^{(1)} = h^{(2)} = h^{(3)} = h^{(4)} \approx 2h^{(5)} \approx 2h^{(6)} \approx \dots \approx 2h^{(12)}$.

The simulation is carried out until the steady state is reached. In this study, the steady state criteria is

$$\frac{\|\mathbf{u}_n - \mathbf{u}_{n-1}\|_\infty}{U} < 5 \times 10^{-7} \quad \text{and} \quad \|p_n - p_{n-1}\|_\infty < 5 \times 10^{-7} \quad (66)$$

◇ Ghia et al. [68] — IMEX-SAV - - - IMEX-SAV with DD Proposed MTS-IMEX-SAV

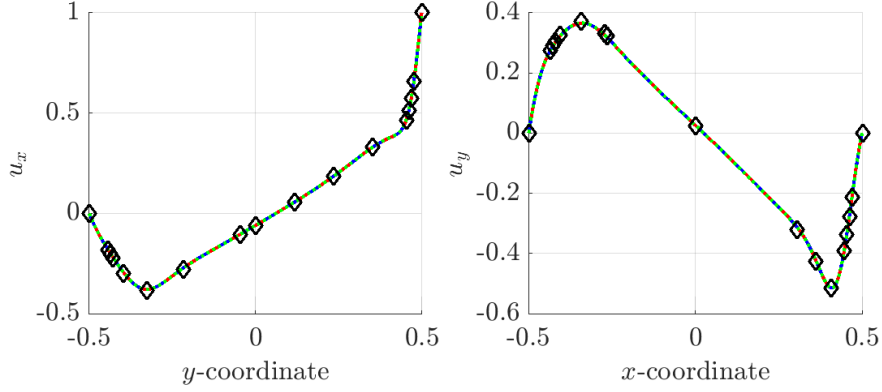


Fig. 9 u_x along $x = 0$ (left) and u_y along $y = 0$ (right) at $Re = 1,000$.

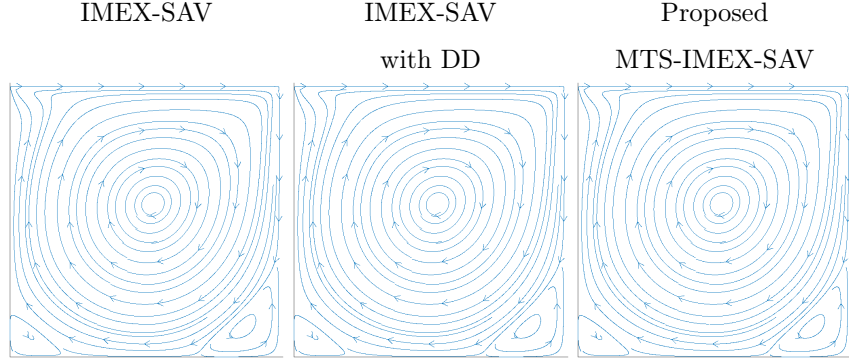


Fig. 10 Streamline pattern using the UTS and MTS methods at $Re = 1,000$.

Fig. 9 and 10 show velocity variations and streamlines, respectively, for $Re = 1,000$. We observe that results from the UTS and MTS methods are similar to Ghia's results. To quantify errors, we define the following relative L^2 error in the steady-state solution: $L^2_{rel}(x) = \|x_N - x_{ref}\|_2 / \|x_{ref}\|_2$, where x is u_x along $x = 0$ or u_y along $y = 0$, and x_{ref} is the solution in [68]. Table 2 shows relative L^2 errors and computational costs when $Re = 100, 400$, and $1,000$. Using the proposed MTS method, we can reduce the computational cost while maintaining similar accuracy. For instance, when $Re = 400$, the proposed MTS-IMEX-SAV method requires roughly 0.19 times the computational costs of the conventional IMEX-SAV method and yields smaller errors. Moreover, it is approximately 1.5 times faster than the IMEX-SAV method with DD.

4.3. Flow around a fixed cylinder

As shown in Fig. 11, we consider 2D incompressible laminar flows around a circular cylinder, which of the center is located at the origin of the coordinate, in a domain $\Omega = [-8D, 18D] \times [-6D, 6D]$, which is divided into 11 subdomains for domain decomposition. All simulations are carried out using 5,850 Q2-Q1 elements at $Re = 40$ and 100 . Here, the Reynolds number is defined as $Re = DU_\infty / \nu$, where D is the diameter of the cylinder and U_∞ is the free-stream velocity. The boundary condition on the cylinder surface is the no-slip

Table 2 Computational costs and relative L^2 errors in u_x along $x = 0$ and u_y along $y = 0$ using the IMEX-SAV and proposed MTS-IMEX-SAV methods for the BDF2 scheme at $Re = 100, 400$, and $1,000$.

Re	Method	Simulation duration, T	Number of time steps, N	Error (%)		Comput. cost (min)	Cost per DOF (ms)
				$L_{rel}^2(u_x)$	$L_{rel}^2(u_y)$		
100	IMEX-SAV	13.38	3,417	0.45	2.74	25.00	33.48
	IMEX-SAV with DD	13.38	3,417	0.45	2.74	7.09	9.49
	Proposed MTS-IMEX-SAV	14.66	1,873	0.45	2.78	4.78	6.40
400	IMEX-SAV	31.63	8,081	0.55	1.27	58.34	78.13
	IMEX-SAV with DD	31.63	8,081	0.55	1.27	16.48	22.07
	Proposed MTS-IMEX-SAV	34.35	4,387	0.53	1.23	10.66	14.28
1,000	IMEX-SAV	44.30	11,317	0.94	1.69	81.32	108.90
	IMEX-SAV with DD	44.30	11,317	0.94	1.69	23.44	31.39
	Proposed MTS-IMEX-SAV	51.47	6,575	0.88	1.64	15.87	21.25

condition: $u_x = u_y = 0$, while the boundary conditions on the walls are the slip condition: $u_y = 0$. A uniform inlet boundary condition is imposed on the inlet: $u_x = U_\infty$ and $u_y = 0$. On the outlet, we impose no tangential stress condition: $\tau_N = 0$.

To simulate this problem, we use the conventional IMEX-SAV method, IMEX-SAV method with DD, and the proposed MTS-IMEX-SAV method for the BDF2 scheme. For the UTS method, we select the time step corresponding to $CFL = 0.4$: $\Delta t = CFL \times h/U_\infty$. On the other hand, in the MTS method, the time step for subdomains $\Omega^{(4)}$ is selected as $\Delta t^{(4)} = \Delta t$. Other time steps are chosen such that $m^{(1)} = 1$, $m^{(2)} = m^{(3)} = 2$, $m^{(4)} = m^{(5)} = \dots = m^{(11)} = 4$.

For steady flows (e.g., $Re = 40$), we measure not only pressure coefficient C_p and drag coefficient C_d but also the nondimensional length of the recirculation x_s/D and location of the center of the vortices $(x_c/D, y_c/D)$ (see Fig. 12). At $Re = 100$ when a periodic vortex street is established, we measure drag

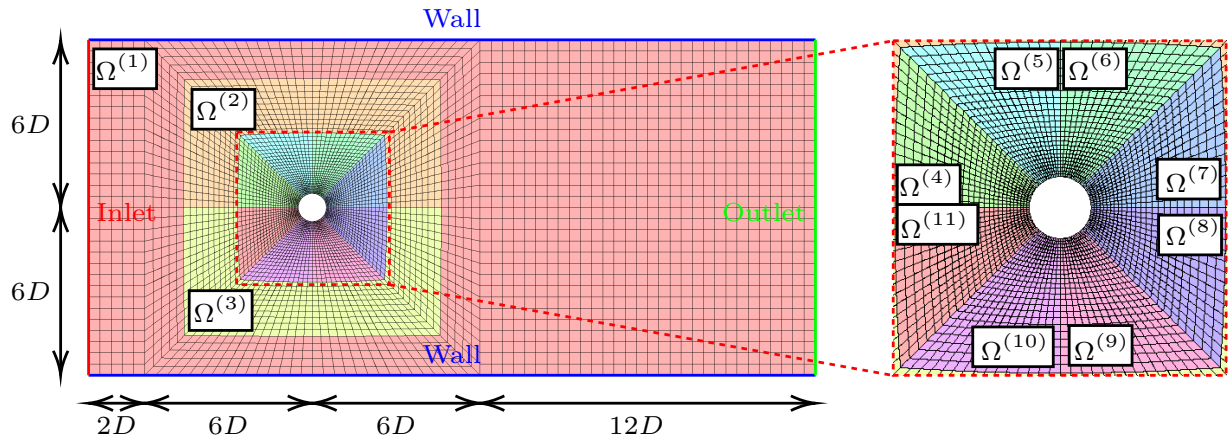


Fig. 11 Geometry and finite element model with 11 subdomains in the cylinder flow problem. The color denotes the subdomain.

441 coefficient C_d , lift coefficient C_l , and Strouhal number $St = fD/U$. The Strouhal frequency f can be defined
 442 as fluctuation frequency of lift coefficient.

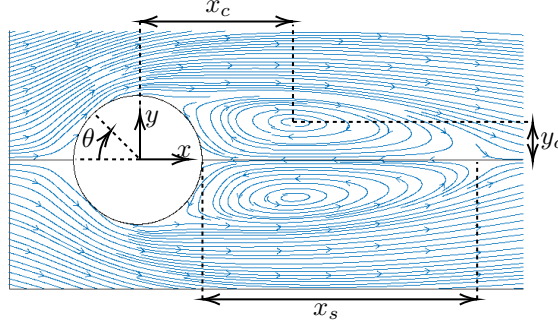


Fig. 12 Streamline pattern at $Re = 40$ using the proposed MTS-IMEX-SAV method. x_s is the length of the recirculation, and (x_c, y_c) is the center of an upper vortex.

443 As seen in Table 3, we compare numerical results using the conventional IMEX-SAV method, IMEX-SAV
 444 method with DD, and proposed MTS-IMEX-SAV method for the BDF2 scheme and several experimental
 445 [69, 70] and numerical [71–74] results of drag coefficients and recirculation characteristics. Fig. 13 shows
 446 variations of pressure coefficient C_p around the cylinder surface at $Re = 40$, compared to other values in
 447 [71, 72]. These results indicate that the proposed MTS-IMEX-SAV method demonstrates similar behavior
 448 as shown in other experimental and numerical studies. Furthermore, it is approximately 10.2 times faster
 449 than the conventional IMEX-SAV method and 3.4 times faster than the IMEX-SAV method with DD.

450 Fig. 14 shows the dimensionless time histories of drag and lift coefficients at $Re = 100$. We observe that
 451 the fully developed flow with periodic vortex streets is reached at $t = 210D/U$. For validation, fluctuating
 452 drag and lift coefficients, and the Strouhal number are compared with previous studies [69, 71–75]. As shown
 453 in Table 4, the present numerical solutions are similar to reference solutions. In the present solutions, the
 454 proposed MTS-IMEX-SAV method accelerates the simulation, being approximately 12.2 times faster than
 455 the conventional IMEX-SAV method and 4.0 times faster than the IMEX-SAV method with DD.

456 4.4. Flow around an oscillating cylinder

457 In this section, instead of the fixed cylinder in Section 4.3, this cylinder is free to vibrate in a cross-flow
 458 direction. The geometry and mesh in this section are the same as those given in Section 4.3. We study fluid
 459 flows on the moving cylinder to test the proposed MTS algorithms for fluid-structure interactions (FSI).
 460 For FSI, one fundamental phenomenon is ‘Lock-in’, or synchronization between vortex shedding frequency
 461 and natural frequency of structures. The lock-in region is defined as the region where the vortex shedding
 462 frequency does not match the expected frequency based on the Reynolds number, but locks on the natural
 463 frequency.

464 The incompressible Navier-Stokes equations are solved in an arbitrary Lagrangian-Eulerian (ALE) for-
 465 mulation [76]. For mesh motion, we divide a problem domain into three non-overlapping regions: Ω_f , Ω_{lm} ,

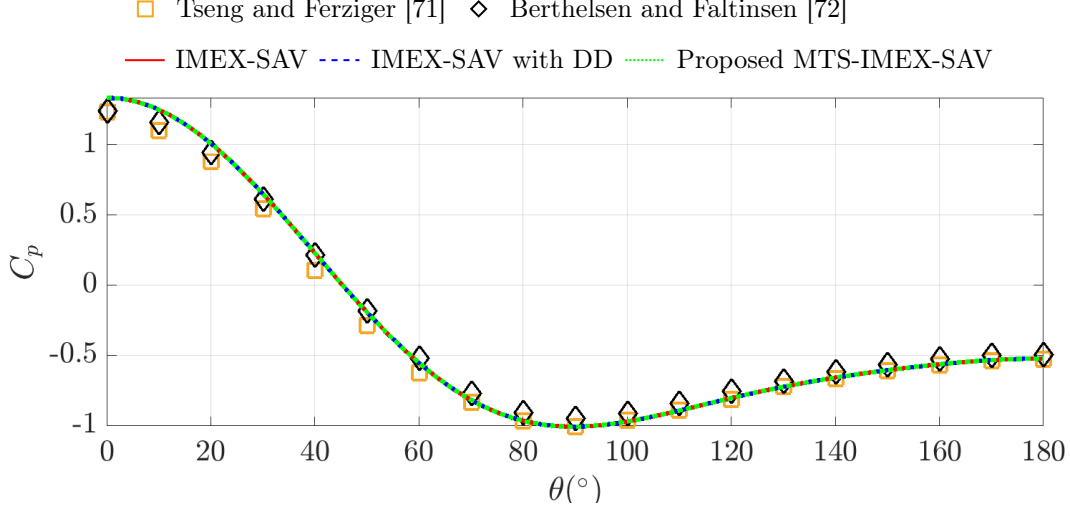


Fig. 13 Variation of pressure coefficients around the cylinder surface at $Re = 40$.

Table 3 Comparison of drag coefficient and recirculation characteristics to other experimental and numerical results at $Re = 40$.

	C_d	x_s/D	x_c/D	y_c/D	Comput. cost	Cost per DOF
Tritton* [69]	1.48	-	-	-		
Coutanceau and Bouard* [70]	-	2.13	1.26	0.29		
Tseng and Ferziger [71]	1.53	2.21	-	-		
Berthelsen and Faltinsen [72]	1.59	2.29	1.22	0.30		
Xu [73]	1.66	2.24	1.22	0.30		
Lepilliez et al. [74]	1.62	2.22	1.21	0.30		
Present study (IMEX-SAV)	1.69	2.19	1.20	0.29	234.67 min	0.2636 sec
Present study (IMEX-SAV with DD)	1.69	2.19	1.20	0.29	79.26 min	0.0890 sec
Present study (Proposed MTS-IMEX-SAV)	1.69	2.19	1.20	0.29	23.12 min	0.0260 sec

* Experimental results

and Ω_{um} , where Ω_f is the region of fixed elements, Ω_{lm} is the region of deformed elements with a Laplacian smoothing moving mesh algorithm, and Ω_{um} is the region of uniform moving meshes. In region Ω_f , mesh displacement vector $\mathbf{d}$ and mesh velocity vector $\mathbf{v}$ are always zero. For all nodes in Ω_{um} , $\mathbf{d} = [0; \delta]$ and $\mathbf{v} = [0; \dot{\delta}]$, where δ denotes the vertical displacement of the moving cylinder. Note that the cylinder moves only in the y -direction. On the other hand, for smoothing moving mesh in Ω_{lm} , we solve the following Laplace's equation to obtain mesh displacement and velocity vectors:

$$\nabla^2 \mathbf{d} = \mathbf{0} \quad , \quad \nabla^2 \mathbf{v} = \mathbf{0} \quad \text{in } \Omega_{lm} \quad (67a)$$

$$\mathbf{d} = \mathbf{0} \quad , \quad \mathbf{v} = \mathbf{0} \quad \text{on } \Omega_{lm} \cap \Omega_f \quad (67b)$$

$$\mathbf{d} = [0; \delta], \quad \mathbf{v} = [0; \dot{\delta}] \quad \text{on } \Omega_{lm} \cap \Omega_{um} \quad (67c)$$

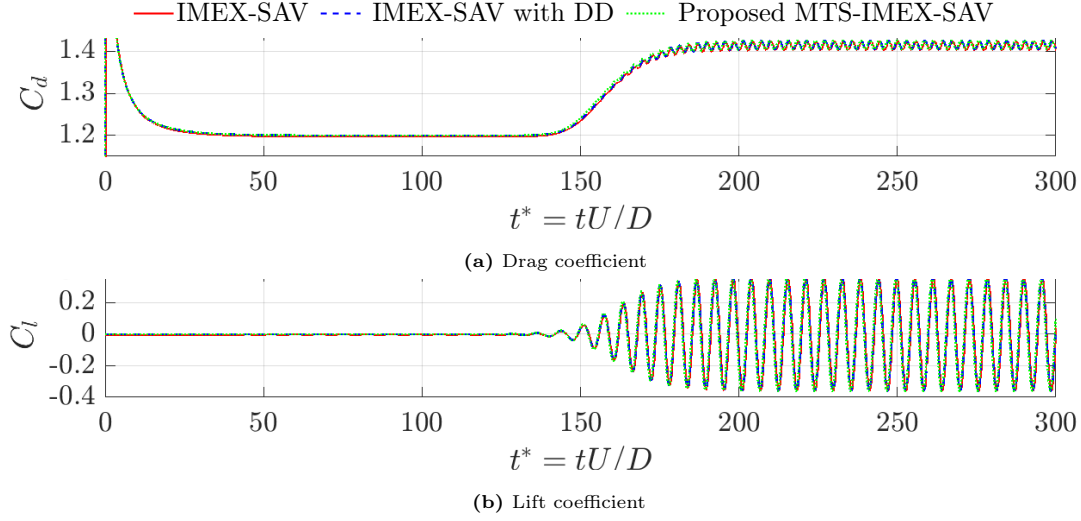


Fig. 14 Drag and lift coefficients at $Re = 100$ using the BDF2 scheme. Note that t^* is dimensionless time.

Table 4 Comparison of drag and lift coefficients, and Strouhal number to other experimental and numerical results at $Re = 100$.

	C_d	C_l	St	Comput. cost	Cost per DOF
Tritton* [69]	1.25	-	-		
Berger and Wille* [75]	-	-	0.16-0.17		
Tseng and Ferziger [71]	1.42	± 0.425	0.164		
Berthelsen and Faltinsen [72]	1.38 ± 0.01	± 0.34	0.169		
Xu [73]	1.42 ± 0.01	± 0.353	0.172		
Lepilliez et al. [74]	1.369 ± 0.01	± 0.321	0.170		
Present study (IMEX-SAV)	1.415 ± 0.01	± 0.359	0.173	19.91 hours	1.3416 sec
Present study (IMEX-SAV with DD)	1.415 ± 0.01	± 0.359	0.173	6.46 hours	0.4353 sec
Present study (Proposed MTS-IMEX-SAV)	1.415 ± 0.01	± 0.360	0.173	1.63 hours	0.1098 sec

* Experimental results

Note that only in Ω_{lm} , we need to update mass matrix $\mathbf{M}^{(s)}$, stiffness matrices $\mathbf{K}^{(s)}$, $\mathbf{K}_p^{(s)}$, and force vector $\mathbf{f}^{(s)}$ (where $\Omega^{(s)} \in \Omega_{lm}$) in every iteration, since $\mathbf{d}$ and $\mathbf{v}$ are uniform throughout Ω_f and Ω_{um} without variation. In this section, we consider that $\Omega_f = \Omega^{(1)}$, $\Omega_{lm} = \Omega^{(2)} \cup \Omega^{(3)}$, and $\Omega_{um} = \Omega^{(4)} \cup \Omega^{(5)} \cup \dots \cup \Omega^{(11)}$.

For boundary conditions, we use the uniform inlet boundary condition on the inlet: $u_x = U_\infty$ and $u_y = 0$, and no tangential stress condition on the outlet: $\tau_N = 0$. We impose a no-slip condition on the circular cylinder surface: $u_x(t) = 0$ and $u_y(t) = \dot{\delta}(t)$. Slip conditions are used on the top and bottom walls: $u_y = 0$.

4.4.1. Forced oscillations

We enforce vertical displacements of a moving cylinder as a sinusoidal function:

$$\delta(t) = \delta_{max} \sin(2\pi f_0 t) \quad (68)$$

where δ_{max} and f_0 are the forced amplitude and frequency, respectively. Koopmann [77] experimented with the effect of forced transverse oscillations of a cylinder on vortex wakes at a low Reynolds number. According to Koopmann's experiments [77], for $A = \delta_{max}/D = 0.25$ at $Re = 100$, the lower and upper bounds of the lock-in region are approximately $F = f_0/f_s = 0.78$ and 1.23, where f_s denotes the vortex shedding frequency of the fixed cylinder. We simulate forced cylinder responses for several frequency ratios $F = 0.5, 0.6, 0.7, 0.75, 0.8, 0.9, 1, 1.1, 1.15, 1.2, 1.25, 1.3$, and 1.4 when $A = 0.25$ and $CFL = 0.4$.

From present numerical results, drag and lift coefficients are compared to the results in [78]. As shown in Fig. 15, there are slight discrepancies, but shapes of drag and lift curves are similar. The gap results from values for a fixed cylinder: $C_d = 1.373 \pm 0.010$ and $C_l = \pm 0.33$ in [78]; and $C_d = 1.415 \pm 0.010$ and $C_l = \pm 0.360$ for present numerical results. Thus, present results are slightly overestimated, compared to those in [78].

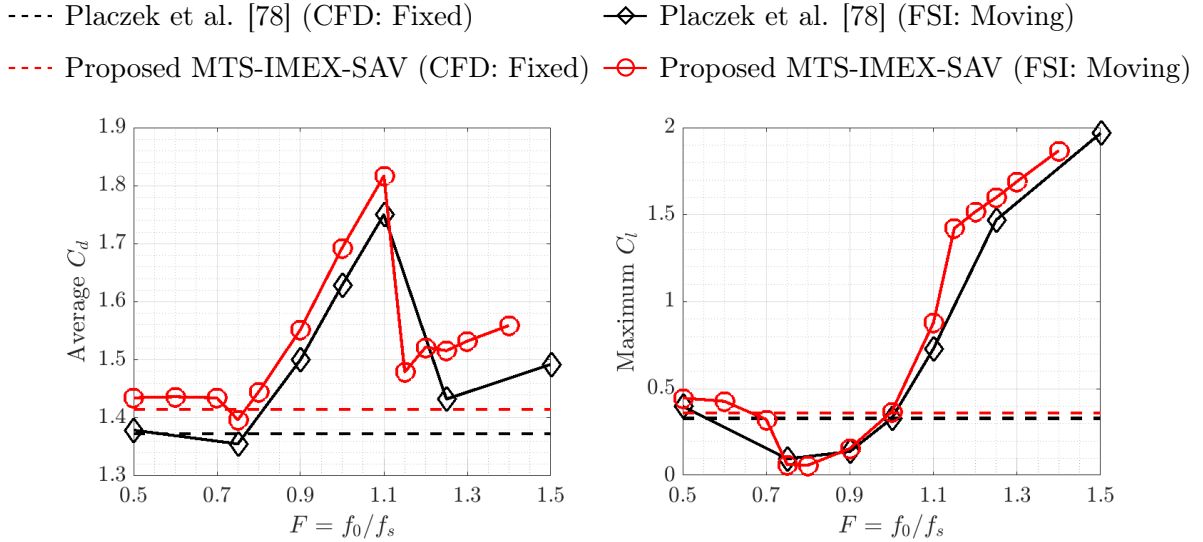


Fig. 15 Average drag coefficients and maximum lift coefficients for various values of F at $Re = 100$.

Fig. 16 shows the power spectral density (PSD) of the lift coefficients for various values of F at $Re = 100$. When $F \leq 0.75$, there are two main vortex shedding frequencies: 1) forced frequency f_0 and 2) vortex shedding frequency for a fixed cylinder f_s . Similarly, when $F \geq 1.2$, we observe two dominant frequencies: f_s and approximately $0.925f_0$, which is similar to the forced frequency. However, when $0.8 \leq F \leq 1.15$, we observe only a primary frequency, which diverges from the vortex shedding frequency for a stationary cylinder and locks in the excited frequency: lock-in configuration.

4.4.2. Vortex-induced vibrations

In this section, an elastically mounted rigid cylinder is simulated in the 2D incompressible flow [79, 80]. The rigid cylinder vibrates only in the cross-flow direction. The equation of motion for the cylinder is

$$m\ddot{\delta}(t) + c\dot{\delta}(t) + k\delta(t) = L(t) \quad (69)$$

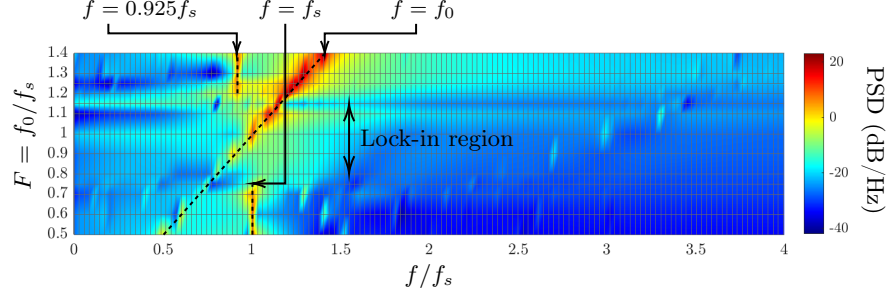


Fig. 16 Power spectral density (PSD) for various values of F at $\text{Re} = 100$. The frequency resolution is $df = 0.005$ (i.e. $df/f_s \approx 0.029$).

where m , c , k , and L are the mass, damping, stiffness, and lift (which is calculated from the fluid part), respectively. As the initial conditions, we use a preliminary solution for the fixed cylinder at $\text{Re} = 100$, as in Section 4.3 (i.e. when a periodic vortex street is established). Structure and fluid properties are the same as those given in [79]: 1) Fluid part: density $\rho = 1000 \text{ kg/m}^3$ and kinematic viscosity $\nu = 0.89 \times 10^{-6} \text{ m}^2/\text{s}$; and 2) Structure part: mass $m = 35.75 \times 10^{-3} \text{ kg}$, damping $c = 0.0039 \text{ kg/s}$, stiffness $k = 69.48 \text{ kg/s}^2$, diameter $D = 1.6 \text{ mm}$, submerged length $l_{\text{sub}} = 120 \text{ mm}$, and natural frequency $f_n = 7.0164 \text{ Hz}$.

To couple the parts of the fluid and the structure, we employ a predictor-corrector algorithm, as shown in Algorithm 1. To compute the predicted displacement and velocity of the structure, second-order extrapolation is used: $\delta_{n+j/m}^{(s)} = (1+j/m)\delta_n - (j/m)\delta_{n-1}$ and $\dot{\delta}_{n+j/m}^{(s)} = (1+j/m)\dot{\delta}_n - (j/m)\dot{\delta}_{n-1}$. For the fluid part, we employ the proposed MTS-IMEX-SAV method for the BDF2 scheme to solve the Navier-Stokes equations in ALE formulation. On the other hand, for the structure part, to solve Eq. (69), we use the IMEX-BDF2-SAV scheme [20] with a large time step of $\Delta t^{(1)}$ to obtain displacement δ_{n+1} and velocity $\dot{\delta}_{n+1}$ at time t_{n+1} .

Algorithm 1 Time-stepping loop at time $t_{n+1} = t_n + \Delta t^{(1)}$ in the predictor-corrector algorithm for FSI simulations.

1. Predictor for structure part

- a) Compute predicted states $\delta_{n+j/m}^{(s)}$ and $\dot{\delta}_{n+j/m}^{(s)}$ using k -point extrapolation (i.e. combination of states at $t_n, t_{n-1}, \dots, t_{n-k+1}$)

2. Moving mesh

- a) Compute mesh displacement $\mathbf{d}_{n+j/m}^{(s)}$ from Eq. (67)
- b) Construct $\mathbf{M}_{n+j/m}^{(s)}$, $\mathbf{K}_{n+j/m}^{(s)}$, $\mathbf{H}_{n+j/m}^{(s)}$, and $\mathbf{f}_{n+j/m}^{(s)}$ at time $t_{n+j/m}^{(s)}$ only in $\Omega^{(s)} \in \Omega_{lm}$

3. Fluid part

- a) Compute mesh velocity $\mathbf{v}_{n+j/m}^{(s)}$ from Eq. (67)
- b) Compute states $\mathbf{x}_{n+j/m}^{(s)}$ by solving Eq. (1) in the ALE formulation
- c) Compute lift force L_{n+1}

4. Corrector for structure part

- a) Update states δ_{n+1} and $\dot{\delta}_{n+1}$ by solving Eq. (69)
-

We carry out a series of computations on vortex-induced vibrations of the cylinder at $Re = 100, 110, 120, 130$, and 140 by increasing the free-stream velocity U . Table 5 shows the comparison of the numerical results for the fixed cylinder (CFD) and the moving cylinder (FSI). Fig. 17 shows the vortex-induced frequencies and amplitude of vertical displacements of the cylinder. For the fixed cylinder, the vortex shedding frequency is linearly increased as an increase in the Reynolds number. On the other hand, for the elastically mounted cylinder, the vortex shedding frequency is locked to the natural frequency f_n . The present predictor-corrector coupled algorithm with the proposed MTS method can capture the lock-in region over the Reynolds number range of 110-130.

Table 5 Drag and lift coefficients, Strouhal number, and vertical displacement of the cylinder at $Re = 100, 110, 120, 130$, and 140 .

Re	CFL	CFD: Fixed cylinder			FSI: Moving cylinder			
		C_d	C_l	St	C_d	C_l	St	δ/D
100	0.4	1.415 ± 0.010	± 0.360	0.173	1.418 ± 0.012	± 0.382	0.173	± 0.005
110	0.4	1.404 ± 0.013	± 0.402	0.180	1.879 ± 0.334	± 0.355	0.183	± 0.375
120	0.35	1.397 ± 0.017	± 0.443	0.183	1.615 ± 0.181	± 0.193	0.168	± 0.301
130	0.35	1.391 ± 0.020	± 0.482	0.187	1.345 ± 0.047	± 0.094	0.155	± 0.170
140	0.35	1.388 ± 0.024	± 0.521	0.190	1.402 ± 0.065	± 0.538	0.190	± 0.054

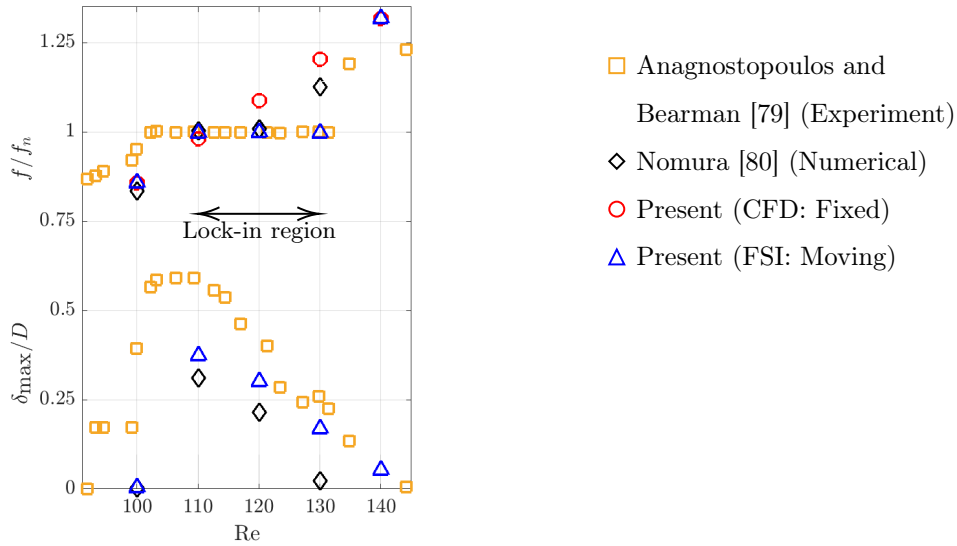


Fig. 17 Vortex-induced frequency f and maximum vertical displacement $\delta_{\max}$ for various values of Re .

5. Concluding remarks

In this paper, we propose a new stable MTS method for coupling multiple IMEX time integration schemes based on SAV stabilization for incompressible Navier-Stokes equations with finite element discretizations. In contrast to conventional IMEX-SAV schemes, the proposed MTS method uses multiple constraint conditions to enforce continuity of velocities and pressures between subdomains, while maintaining overall stability through SAVs from individual subdomains. We use linear interpolation to compute Lagrange multipliers at subdomain time steps and a global SAV based on the summation of energies in each subdomain to ensure unconditional stability.

In Section 3.2, we prove that the proposed MTS-IMEX-SAV method is unconditionally stable. Moreover, from accuracy tests (see Section 3.4), we show that the proposed MTS-IMEX-SAV method for the BDF1 and BDF2 schemes preserves the convergence rates (i.e. first- and second-order accuracy, respectively). Note that even though different BDF k schemes may be used in different subdomains, the overall convergence rate is governed by the minimum among all subdomains.

To demonstrate the performance of the proposed MTS-IMEX-SAV method, we simulate several numerical examples. From results in Sections 4.1 and 4.2, we show that compared to errors at a similar computational effort, the proposed MTS method yields lower errors in velocity and pressure. From results of 2D cylinder flows (see Section 4.3), the numerical results are compared to several references, and we notice that the proposed MTS method yields very similar results with lower computational costs compared to the IMEX-SAV methods. Lastly, in Section 4.4, we implement the proposed MTS method into a predictor-corrector FSI coupling algorithm to show its performance for simulating FSI problems.

In this study, the proposed MTS-IMEX-SAV method is used to solve incompressible Navier-Stokes equations in the coupled velocity-pressure formulation. In future, the proposed MTS method may be modified to implement projection methods, such as pressure-correction; velocity-correction; and consistent splitting techniques, to be more efficient. In addition, the proposed MTS method may be extended to compressible Navier-Stokes equations as well.

CRedit authorship contribution statement

Sun-Beom Kwon: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Arun Prakash:** Conceptualization, Funding acquisition, Project administration, Resources, Supervision, Writing – original draft, Writing – review & editing.

Data availability

Data may be available upon request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A Collected block matrices $\mathbb{B}^I$ and $\mathbb{Y}^I$ for the interface problem.

$$\mathbb{B}^I = \begin{bmatrix} \mathbb{B}^{(1,2)} & \mathbb{B}^{(2,1)} & & & & \\ & \vdots & & \ddots & & \\ & \mathbb{B}^{(1,s)} & & & \mathbb{B}^{(s,1)} & \\ \mathbb{B}^{(1,s+1)} & & & & \mathbb{B}^{(s+1,1)} & \\ & \vdots & & & & \ddots \\ \mathbb{B}^{(1,S-1)} & & & & & \mathbb{B}^{(S-1,1)} \\ \mathbb{B}^{(1,S)} & & & & & \mathbb{B}^{(S,1)} \\ \hline & & \ddots & & & \vdots \\ \hline & & & \mathbb{B}^{(s,s+1)} & \mathbb{B}^{(s+1,s)} & \\ & & & \vdots & & \ddots \\ & & & \mathbb{B}^{(s,S-1)} & & \mathbb{B}^{(S-1,s)} \\ & & & \mathbb{B}^{(s,S)} & & \mathbb{B}^{(S,s)} \\ \hline & & & & \ddots & \vdots \\ \hline & & & & & \mathbb{B}^{(S-1,S)} & \mathbb{B}^{(S,S-1)} \end{bmatrix} \quad (\text{A.1})$$

$$\mathbb{Y}^I = \left[\begin{array}{cccccc|cccc|cc|cc} \mathbb{Y}_1^{(1,2)} & \dots & \mathbb{Y}_1^{(1,s)} & \mathbb{Y}_1^{(1,s+1)} & \dots & \mathbb{Y}_1^{(1,S-1)} & \mathbb{Y}_1^{(1,S)} & & & & & \\ \mathbb{Y}_1^{(2,1)} & & & & & & & \ddots & & & & \\ & \ddots & & & & & & & & & & \\ & & \mathbb{Y}_1^{(s,1)} & & & & & & & & & \\ & & & \mathbb{Y}_1^{(s+1,1)} & & & & & & & & \\ & & & & \ddots & & & & & & & \\ & & & & & \mathbb{Y}_1^{(S-1,1)} & & & & & & \\ & & & & & & \mathbb{Y}_1^{(S,1)} & \dots & & & & \\ \hline & & & & & & & & & & & \\ \hline & & & & & & & & \mathbb{Y}_1^{(s,s+1)} & \dots & \mathbb{Y}_1^{(s,S-1)} & \mathbb{Y}_1^{(s,S)} \\ & & & & & & & & \mathbb{Y}_1^{(s+1,s)} & & & \ddots \\ & & & & & & & & & \ddots & & \\ & & & & & & & & & & \mathbb{Y}_1^{(S-1,s)} & \\ & & & & & & & & & & \mathbb{Y}_1^{(S,s)} & \dots \\ & & & & & & & & & & & \mathbb{Y}_1^{(S-1,S)} \\ & & & & & & & & & & & \mathbb{Y}_1^{(S,S-1)} \end{array} \right] \quad (\text{A.2})$$

References

- [1] T. Chung, Computational fluid dynamics, Cambridge university press, 2010.
- [2] J. Droniou, R. Eymard, Study of the mixed finite volume method for Stokes and Navier-Stokes equations, Numerical Methods for Partial Differential Equations: An International Journal 25 (2009) 137–71.
- [3] O. C. Zienkiewicz, R. L. Taylor, J. Z. Zhu, The finite element method: its basis and fundamentals, Elsevier, 2005.
- [4] G. Giorgiani, S. Fernández-Méndez, A. Huerta, Hybridizable discontinuous Galerkin with degree adaptivity for the incompressible Navier–Stokes equations, Computers & Fluids 98 (2014) 196–208.
- [5] N. Fehn, W. A. Wall, M. Kronbichler, On the stability of projection methods for the incompressible Navier–Stokes equations based on high-order discontinuous Galerkin discretizations, Journal of Computational Physics 351 (2017) 392–421.
- [6] G. Orlando, A. Della Rocca, P. F. Barbante, L. Bonaventura, N. Parolini, An efficient and accurate implicit DG solver for the incompressible Navier–Stokes equations, International Journal for Numerical Methods in Fluids 94 (2022) 1484–516.
- [7] G. Wanner, E. Hairer, Solving ordinary differential equations II, volume 375, Springer Berlin Heidelberg New York, 1996.
- [8] J. N. Reddy, D. K. Gartling, The finite element method in heat transfer and fluid dynamics, CRC press, 2010.
- [9] F. Bashforth, J. C. Adams, An attempt to test the theories of capillary action by comparing the theoretical and measured forms of drops of fluid, University Press, 1883.
- [10] J. G. Verwer, Explicit Runge-Kutta methods for parabolic partial differential equations, Applied Numerical Mathematics 22 (1996) 359–79.
- [11] J. Crank, P. Nicolson, A practical method for numerical evaluation of solutions of partial differential equations of the heat-conduction type, in: Mathematical proceedings of the Cambridge philosophical society, volume 43, Cambridge University Press, 1947, pp. 50–67.
- [12] F. R. Moulton, New methods in exterior ballistics, University of Chicago Press, 1926.
- [13] C. Gear, Numerical initial value problems in ordinary differential equations, 1971.
- [14] J. C. Butcher, A history of Runge-Kutta methods, Applied numerical mathematics 20 (1996) 247–60.
- [15] J. C. Butcher, Numerical methods for ordinary differential equations in the 20th century, Journal of Computational and Applied Mathematics 125 (2000) 1–29.

- [16] K. E. Jansen, C. H. Whiting, G. M. Hulbert, A generalized- α method for integrating the filtered Navier–Stokes equations with a stabilized finite element method, *Computer Methods in Applied Mechanics and Engineering* 190 (2000) 305–19.
- [17] K. Tamma, S. U. Masuri, A novel extension of GS4-1 time integrator to fluid dynamics type non-linear problems with illustrations to Burgers’ equation, *International Journal of Numerical Methods for Heat & Fluid Flow* 26 (2016) 1634–60.
- [18] J. C. Butcher, *Numerical methods for ordinary differential equations*, John Wiley & Sons, 2016.
- [19] R. Courant, K. Friedrichs, H. Lewy, Über die partiellen differenzengleichungen der mathematischen physik, *Mathematische annalen* 100 (1928) 32–74.
- [20] S.-B. Kwon, A. Prakash, Scalar auxiliary variable (SAV) stabilization of implicit-explicit (IMEX) time integration schemes for non-linear structural dynamics, *International Journal of Numerical Methods in Engineering* 126 (2025) e7660.
- [21] M. Dumbser, V. Casulli, A conservative, weakly nonlinear semi-implicit finite volume scheme for the compressible Navier–Stokes equations with general equation of state, *Applied Mathematics and Computation* 272 (2016) 479–97.
- [22] M. Tan, J. Cheng, C.-W. Shu, Stability of high order finite difference and local discontinuous galerkin schemes with explicit-implicit-null time-marching for high order dissipative and dispersive equations, *Journal of Computational Physics* 464 (2022) 111314.
- [23] J. Shen, J. Xu, J. Yang, The scalar auxiliary variable (SAV) approach for gradient flows, *Journal of Computational Physics* 353 (2018) 407–16.
- [24] J. Shen, J. Xu, J. Yang, A new class of efficient and robust energy stable schemes for gradient flows, *SIAM Review* 61 (2019) 474–506.
- [25] F. Huang, J. Shen, Z. Yang, A highly efficient and accurate new scalar auxiliary variable approach for gradient flows, *SIAM Journal on Scientific Computing* 42 (2020) A2514–36.
- [26] F. Huang, J. Shen, A new class of implicit–explicit BDFk SAV schemes for general dissipative systems and their error analysis, *Computer Methods in Applied Mechanics and Engineering* 392 (2022) 114718.
- [27] L. Lin, Z. Yang, S. Dong, Numerical approximation of incompressible Navier-Stokes equations based on an auxiliary energy variable, *Journal of Computational Physics* 388 (2019) 1–22.
- [28] X. Li, J. Shen, Error analysis of the SAV-MAC scheme for the Navier-Stokes equations, *SIAM Journal on Numerical Analysis* 58 (2020) 2465–91.

- [29] F. Huang, J. Shen, Stability and error analysis of a class of high-order IMEX schemes for Navier–Stokes equations with periodic boundary conditions, *SIAM Journal on Numerical Analysis* 59 (2021) 2926–54.
- [30] X. Li, J. Shen, Z. Liu, New SAV-pressure correction methods for the Navier-Stokes equations: stability and error analysis, *Mathematics of Computation* 91 (2022) 141–67.
- [31] K. Wu, F. Huang, J. Shen, A new class of higher-order decoupled schemes for the incompressible Navier-Stokes equations and applications to rotating dynamics, *Journal of Computational Physics* 458 (2022) 111097.
- [32] T. Belytschko, W. K. Liu, P. Smolinski, Multi-stepping implicit-explicit procedures in transient analysis, in: *Innovative methods in Nonlinear Analysis*, Pineridge Press, 1984, pp. 135–54.
- [33] W. Daniel, The subcycled Newmark algorithm, *Computational Mechanics* 20 (1997) 272–81.
- [34] A. Gravouil, A. Combescure, Multi-time-step explicit–implicit method for non-linear structural dynamics, *International Journal for Numerical Methods in Engineering* 50 (2001) 199–225.
- [35] A. Prakash, K. D. Hjelmstad, A FETI-based multi-time-step coupling method for Newmark schemes in structural dynamics, *International Journal for Numerical Methods in Engineering* 61 (2004) 2183–204.
- [36] N. Mahjoubi, A. Gravouil, A. Combescure, N. Greffet, A monolithic energy conserving method to couple heterogeneous time integrators with incompatible time steps in structural dynamics, *Computer Methods in Applied Mechanics and Engineering* 200 (2011) 1069–86.
- [37] M. Brun, A. Gravouil, A. Combescure, A. Limam, Two FETI-based heterogeneous time step coupling methods for Newmark and α -schemes derived from the energy method, *Computer methods in applied mechanics and engineering* 283 (2015) 130–76.
- [38] A. Gravouil, A. Combescure, M. Brun, Heterogeneous asynchronous time integrators for computational structural dynamics, *International Journal for Numerical Methods in Engineering* 102 (2015) 202–32.
- [39] D. Tae, K. K. Tamma, A robust and generalized effective DAE framework encompassing different methods, algorithms, and model order reduction for linear and nonlinear second order dynamical systems, *Finite Elements in Analysis and Design* 228 (2024) 104043.
- [40] S.-B. Kwon, A. Prakash, Coupling composite schemes with different time steps for multi-scale structural dynamics, *International Journal for Multiscale Computational Engineering* 23 (2025) 63–109.
- [41] S.-B. Kwon, A. Prakash, A unified computational framework for developing general multi-time-step (MTS) coupling schemes for time-dependent problems, *Engineering with Computers* 41 (2025) 4727–56.

- [42] K. B. Nakshatrala, K. D. Hjelmstad, D. A. Tortorelli, A FETI-based domain decomposition technique for time-dependent first-order systems based on a DAE approach, *International Journal for Numerical Methods in Engineering* 75 (2008) 1385–415.
- [43] K. Nakshatrala, A. Prakash, K. Hjelmstad, On dual Schur domain decomposition method for linear first-order transient problems, *Journal of Computational Physics* 228 (2009) 7957–85.
- [44] P. B. Nakshatrala, K. B. Nakshatrala, D. A. Tortorelli, A time-staggered partitioned coupling algorithm for transient heat conduction, *International Journal for Numerical Methods in Engineering* 78 (2009) 1387–406.
- [45] S. Osher, R. Sanders, Numerical approximations to nonlinear conservation laws with locally varying time and space grids, *Mathematics of computation* 41 (1983) 321–36.
- [46] C. J. Trahan, C. Dawson, Local time-stepping in Runge–Kutta discontinuous Galerkin finite element methods applied to the shallow-water equations, *Computer Methods in Applied Mechanics and Engineering* 217 (2012) 139–52.
- [47] M. Emmett, W. Zhang, J. B. Bell, High-order algorithms for compressible reacting flow with complex chemistry, *Combustion Theory and Modelling* 18 (2014) 361–87.
- [48] I. Rybak, J. Magiera, R. Helmig, C. Rohde, Multirate time integration for coupled saturated/unsaturated porous medium and free flow systems, *Computational Geosciences* 19 (2015) 299–309.
- [49] S. Gupta, B. Wohlmuth, R. Helmig, Multi-rate time stepping schemes for hydro-geomechanical model for subsurface methane hydrate reservoirs, *Advances in Water Resources* 91 (2016) 78–87.
- [50] C. Mikida, A. Klöckner, D. Bodony, Multi-rate time integration on overset meshes, *Journal of Computational Physics* 396 (2019) 325–46.
- [51] K. Mittal, S. Dutta, P. Fischer, Multirate timestepping for the incompressible Navier-Stokes equations in overlapping grids, *Journal of Computational Physics* 437 (2021) 110335.
- [52] J. Li, A dual-primal FETI method for solving Stokes/Navier-Stokes equations, in: *Proceedings of the 14th International Conference on Domain Decomposition Methods in Science and Engineering*, Cocoyoc, Mexico, 2003, pp. 225–31. Available online at www.ddm.org/DD14.
- [53] J. Li, A dual-primal FETI method for incompressible Stokes equations, *Numerische Mathematik* 102 (2005) 257–75.
- [54] H. H. Kim, C.-O. Lee, E.-H. Park, A FETI-DP formulation for the Stokes problem without primal pressure components, *SIAM Journal on Numerical Analysis* 47 (2010) 4142–62.

- [55] H. Benhassine, A. Bendali, A non-overlapping domain decomposition method for continuous-pressure mixedfinite element approximations of the Stokes problem, *ESAIM: Mathematical Modelling and Numerical Analysis* 45 (2011) 675–96.
- [56] J. Li, X. Tu, A nonoverlapping domain decomposition method for incompressible Stokes equations with continuous pressures, *SIAM Journal on Numerical Analysis* 51 (2013) 1235–53.
- [57] X. Tu, J. Li, A FETI-DP type domain decomposition algorithm for three-dimensional incompressible Stokes equations, *SIAM Journal on Numerical Analysis* 53 (2015) 720–42.
- [58] C. Farhat, M. Lesoinne, Two efficient staggered algorithms for the serial and parallel solution of three-dimensional nonlinear transient aeroelastic problems, *Computer methods in applied mechanics and engineering* 182 (2000) 499–515.
- [59] C. A. Felippa, K.-C. Park, C. Farhat, Partitioned analysis of coupled mechanical systems, *Computer methods in applied mechanics and engineering* 190 (2001) 3247–70.
- [60] K.-J. Bathe, *Finite element procedures*. 2nd ed, Higher Education Press China, Watertown (MA), 2016.
- [61] C. Taylor, P. Hood, A numerical solution of the Navier-Stokes equations using the finite element technique, *Computers & Fluids* 1 (1973) 73–100.
- [62] X. Wang, A. Prakash, J.-S. Chen, E. Taciroglu, Variationally consistent coupling of non-matching discretizations for large deformation problems, *Computational Mechanics* 60 (2017) 465–78.
- [63] M. H. Jamal, A. Prakash, M. Kulkarni, Exploiting semantics of temporal multi-scale methods to optimize multi-level mesh partitioning, *International Journal for Numerical Methods in Engineering* 112 (2017) 58–85.
- [64] T. Zhang, J. Yuan, Unconditional stability and optimal error estimates of Euler implicit/explicit-SAV scheme for the Navier-Stokes equations, *Journal of Scientific Computing* 90 (2022) 1–20.
- [65] D. S. Clark, Short proof of a discrete Gronwall inequality, *Discrete applied mathematics* 16 (1987) 279–81.
- [66] S. E. Rogers, D. Kwak, Upwind differencing scheme for the time-accurate incompressible Navier-Stokes equations, *AIAA journal* 28 (1990) 253–62.
- [67] A. Shah, L. Yuan, S. Islam, Numerical solution of unsteady Navier–Stokes equations on curvilinear meshes, *Computers & Mathematics with Applications* 63 (2012) 1548–56.
- [68] U. Ghia, K. N. Ghia, C. Shin, High-re solutions for incompressible flow using the Navier-Stokes equations and a multigrid method, *Journal of computational physics* 48 (1982) 387–411.

- [69] D. J. Tritton, Experiments on the flow past a circular cylinder at low Reynolds numbers, *Journal of Fluid Mechanics* 6 (1959) 547–67.
- [70] M. Coutanceau, R. Bouard, Experimental determination of the main features of the viscous flow in the wake of a circular cylinder in uniform translation. Part 1. Steady flow, *Journal of Fluid Mechanics* 79 (1977) 231–56.
- [71] Y.-H. Tseng, J. H. Ferziger, A ghost-cell immersed boundary method for flow in complex geometry, *Journal of computational physics* 192 (2003) 593–623.
- [72] P. A. Berthelsen, O. M. Faltinsen, A local directional ghost cell approach for incompressible viscous flow problems with irregular boundaries, *Journal of computational physics* 227 (2008) 4354–97.
- [73] S. Xu, The immersed interface method for simulating prescribed motion of rigid objects in an incompressible viscous flow, *Journal of Computational Physics* 227 (2008) 5045–71.
- [74] M. Lepilliez, E. R. Popescu, F. Gibou, S. Tanguy, On two-phase flow solvers in irregular domains with contact line, *Journal of Computational Physics* 321 (2016) 1217–51.
- [75] E. Berger, R. Wille, Periodic flow phenomena, *Annual Review of Fluid Mechanics* 4 (1972) 313–40.
- [76] J. Donea, A. Huerta, J.-P. Ponthot, A. Rodríguez-Ferran, Arbitrary Lagrangian–Eulerian methods, *Encyclopedia of Computational Mechanics Second Edition* (2017) 1–23.
- [77] G. Koopmann, The vortex wakes of vibrating cylinders at low Reynolds numbers, *Journal of Fluid Mechanics* 28 (1967) 501–12.
- [78] A. Placzek, J.-F. Sigrist, A. Hamdouni, Numerical simulation of an oscillating cylinder in a cross-flow at low Reynolds number: Forced and free oscillations, *Computers & Fluids* 38 (2009) 80–100.
- [79] P. Anagnostopoulos, P. Bearman, Response characteristics of a vortex-excited cylinder at low Reynolds numbers, *Journal of Fluids and Structures* 6 (1992) 39–50.
- [80] T. Nomura, ALE finite element computations of fluid-structure interaction problems, *Computer methods in applied mechanics and engineering* 112 (1994) 291–308.